SFDA-OCDet: Spatial-Frequency Difference Attention Network for Remote Sensing Object-Level Change Detection

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***Abstract*—Remote sensing image change detection refers to the automatic acquisition of the location and category of object changes using images of different phases. The rapid development of optical remote sensing and artificial intelligence technology has made deep learning a mainstream method for remote sensing change detection. In current research, deep learning-based remote sensing change detection mainly focuses on determining whether changes have occurred at the pixel level, and this pixel-level change detection is susceptible to pseudo-changes due to factors such as differences in imaging angles of optical cameras, changes in imaging seasons, and occlusion. In addition, in some specific scenarios, such as the change analysis task for movable objects with regular geometries (e.g., airplane and vehicles) is more concerned with the change of the object as a whole than the change of the pixels inside the object, in which case pixel-level methods are no longer the best choice and object-level change detection methods should be adopted. To address the above problems, we propose a spatial-frequency difference attention network for object-level change detection (SFDA-OCDet). First, we use a Siamese backbone with shared weight parameters to extract the image features of different stages from the input dual-temporal images. Second, we design the spatial difference attention (SDA) module, which obtains the attention weights from the local context and the global context to enhance the learning ability of the difference features of different time images from the spatial domain. Then, we propose the wavelet frequency difference attention (WFDA) module, which utilizes the discrete wavelet transform (DWT) and the attention mechanism to enhance the difference feature learning capability from the frequency domain. Finally, we present the wavelet balanced feature pyramid (WBFP) module, which introduces the DWT and its inverse transformation to implement the downsampling and upsampling process to avoid feature information loss. The change detection head outputs change detection results in the form of a bounding box. We construct four object-level change detection datasets, Airplane-CD, Vehicle-CD, Building-CD, and Wind turbine-CD, to carry out experimental validation. Numerous experiments have shown that the proposed SFDA-OCDet outperforms other comparative methods, proving the validity of the method.**

***Index Terms*—Attention mechanism, discrete wavelet transform (DWT), object-level change detection, spatial-frequency domain.**

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I. Introduction

THE development of remote sensing technology and artificial intelligence technology has led to important breakthroughs in the fields of remote sensing image change detection [1], object detection and tracking [2], [3], [4], [5], scene classification [6], hyperspectral classification [7], [8], [9], and multimodal analysis [10]. Among them, change detection plays an important role in analyzing the changes in the target at two different moments. The purpose of remote sensing image change detection is to detect the location of a target change in images at different times in the same geographical range [11], [12]. The abundance of high-resolution image data and the rapid development of theoretical methods make change detection widely used in application areas such as land cover monitoring [13], [14], [15], disaster assessment [16], [17], [18], urban planning and management [19], [20], [21], and change analysis of movable objects [22], [23], [24].

Change detection methods can be classified into two types, pixel-level and object-level, based on the division of the type of basic unit of processing. Pixel-level change detection uses the pixel as the basic discriminating unit, and assigns two kinds of category labels with change or nonchange pixel by pixel. Traditional pixel-level change detection methods mainly include image algebraic methods, image transformation methods, and classification methods. The basic idea of the image algebraic method is to directly perform algebraic calculations, such as difference and division, on the pixel values of the corresponding bands in the dual-temporal images to generate a difference image characterizing the changes, and then set an appropriate threshold to obtain the change detection results.

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Commonly used algebraic algorithms include image difference [25], [26], image rationing [27], image regression [28], [29], and change vector analysis (CVA) [30], [31]. The operation of the algebraic method is relatively simple, and it hinges on how to set the appropriate threshold to capture the region of change. The main idea of the image transformation method is to remove the redundant information between dual-temporal images and highlight the change information of the features by processing the images, such as dimension reduction. Typical image transformation methods include principal component analysis (PCA) [32], [33], tasseled cap transformation [34], [35], multivariate alteration detection (MAD) [36], [37], and Gram–Schmidt transformation [38], [39]. The image transformation method may lose feature information associated with changes, and the change detection results lack information about the type of object. The basic process of the classification method is to obtain the classification results of the objects in the dual-temporal images separately and then compare them to get the location and category information of the changes, or to combine the dual-temporal images and then directly classify them to get the change results. In short, there are two types of classification methods: post-classification comparison method [40], [41] and direct classification method [42], [43]. The accuracy of classification-based change detection is related to the performance of the classifiers, thus placing high demands on the classifiers. Traditional pixel-level change detection methods rely on expert experience to design suitable features and set effective thresholds. The detection ability of traditional methods is relatively weak due to the poor representation of manually designed features and the limited performance of classifiers. Deep learning methods have the advantages of outstanding feature representation capability, powerful feature extraction capability at different levels, such as low-level detailed features and high-level semantic features, excellent classifier performance, and applicability to complex scene change detection tasks, which has become the mainstream technical method of change detection and has been widely researched and used. Pixel-level change detection based on deep learning can be considered as a dense prediction problem, in which dual-temporal images are fed into a fully convolutional neural network to determine the class of each pixel through feature extraction, feature recovery, and classification processes, and the class is divided into two categories: change and nonchange. The pixel-level change detection networks based on deep learning can be categorized into two architectures: single-branch and dual-branch [44], [45]. The basic operation of a single-branch network is to fuse two images at different times into a single image by addition or concatenation, and then feed it into a fully convolutional neural network to predict the results of change detection. The basic process of a dual-branch network is that a pair of images is fed into two backbone networks of the same structure to extract the dual-temporal image features, which are processed in difference or concatenation and then fed into a decoding network to predict the change map.

In contrast to pixel-level change detection that uses a single pixel as the unit of analysis, object-level change detection utilizes an object as the basic unit of discrimination. The object contains two forms. One is a region consisting of a cluster of pixels with similar spectral, shape, texture, and other features and distributed adjacent to each other in spatial location [46], [47], which is generally represented by superpixels. The other is an instance that can represent a complete target [48], represented by a bounding box. Traditional object-level change detection utilizes a multiscale segmentation method, taking the maximum heterogeneity between the segmented objects and the maximum homogeneity of the pixels within the object as the basic principle, segmenting the image with the help of spectral, shape, texture, and other feature information to obtain a series of objects represented by superpixels, and then obtaining the change results using methods such as the direct comparison method, the post-classification comparison method, and so on. The basic principle of the direct comparison method is to directly compare the feature information between objects of dual-temporal images to obtain the change detection results. The post-classification comparison method performs object-oriented classification of dual-temporal images and compares the classification results to produce a change map. The key to traditional object-level change detection is image segmentation, the accuracy of which affects the effectiveness of classification and change detection. In addition, the post-classification comparison method places high demands on the performance of object-oriented classifiers. Depending on the form of object representation, object-level change detection methods based on deep learning are classified into two categories. For objects in the form of superpixels, the basic idea is to generate superpixels using multiscale segmentation, and then use models such as stacked autoencoders to learn the changed features between pairs of superpixels and to obtain the results of change [49], [50]. For objects in the form of bounding boxes, the basic idea is to use a deep learning-based object detection network as the infrastructure to generate candidate boxes containing changed objects, and then perform category prediction and bounding box regression on the changed features within the candidate boxes.

Pixel-level change detection requires comparing the change information of pixels at the same location in images at different times. Differences of objects in images with different imaging angles, seasonal changes, and the effects of factors such as occlusion can lead to the problem of pseudo-change in pixel-level change detection results. Object-level change detection in the form of a bounding box can not only alleviate the problem of pseudo-change, but also avoid the over-segmentation and under-segmentation problems due to inappropriate scale parameter settings in the form of superpixel object-level change detection, and it is also suitable for application scenarios that focus on the overall change of the target instead of the change of the internal pixels. To this end, we propose a spatial-frequency difference attention network for remote sensing object-level change detection (SFDA-OCDet). The main contributions of this article can be summarized as follows.

1) We construct an end-to-end object-level change detection network called SFDA-OCDet that achieves superior change detection performance on the four datasets we

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created, Airplane-CD, Vehicle-CD, Building-CD, and Wind turbine-CD.
2) We design the spatial difference attention (SDA) module with the aim of enhancing the model's learning capability of difference features of changed objects in dual-temporal images in the spatial domain.
3) We present the wavelet frequency difference attention (WFDA) module, which uses the discrete wavelet transform (DWT) and attention mechanism to enhance the learning ability of difference features in the frequency domain.
4) We propose the wavelet balanced feature pyramid (WBFP) module, which uses the DWT and its inverse to replace the max pooling and interpolation operations in the BFP module for downsampling and upsampling to reduce the feature information loss.

This article is organized as follows. Section II describes related work. Section III presents the proposed SFDA-OCDet in detail. Section IV reports experimental results and analyses on four object-level change detection datasets. Section V concludes this article.

II. Related Work

A. Object-Level Change Detection Based on Deep Learning

Wang et al. [51] design a remote sensing change detection method based on faster R-CNN for new buildings, roads, and other targets. According to the difference in the way the images are fed into the network, the article designs two networks, MFRCNN and SFRCNN, where the fused images of the two times and the difference image obtained by subtraction are fed into the Faster R-CNN, respectively. The post-processing stage uses the snakes model to generate change detection results in the form of segmentation from the predicted bounding box. Han et al. [22] design a two-stage dual regions of interest networks (DRoINs) to detect changes in objects such as vehicles in aerial images at different times. DRoINs consists of three sub-networks, namely, feature extracted network, change proposal network, and difference judgment network, which adopts a dual-branch network architecture where the three sub-networks share weight parameters in two branches. The processing flow of DRoINs is to extract the features of two images at different times, then use a change proposal network to generate candidate proposals for each image and compare them to get the change proposals. Finally, the difference judgment network is used to judge and analyze whether the features in each change candidate proposal have changed or not, and the output of the network is the location of the changed object represented by the bounding box. Zhang et al. [48] propose a one-stage object-level change detection network (DCA-Det) to detect changes of artificial objects such as buildings in remote sensing images. DCA-Det first extracts the features in dual-temporal images using a weight-sharing Siamese backbone, and designs a location-correlated attention module and a channel-correlated attention module to obtain multiscale difference features, then fuses them at different scales using a refined PAFPN, and finally predicts the location and confidence of the changed object. Zhang et al. [52] propose an object-level change detection model based on Siamese YOLOv5, which also uses a Siamese backbone and then exploits a structural similarity-based metric to compute the difference features, and finally inputs the difference features into the YOLOv5 detector to predict the change results. Huo et al. [53] design a one-stage object-level change detection method for natural scenes called SCTF-Det. The basic idea of this method is to first extract image features by using a Siamese backbone composed of ResNet [54] and vision Transformer [55], and then cascade the difference features obtained by alternately subtracting dual-temporal features from the original features, and finally feed the fused features into CenterNet [56] to produce the prediction results.

B. Application of Attention Mechanism

The purpose of applying the attention mechanism to an object-level or pixel-level change detection task is to enhance the features of changed regions and attenuate the features of unchanged regions. In the existing studies, the attention mechanism in change detection is mainly based on the spatial attention and channel attention in the spatial domain, which realizes the enhancement of feature information by establishing the correlation between different spatial locations of features and the correlation between different channels, respectively.

The AGCDetNet proposed by Song and Jiang [57] builds a spatial attention module to improve the distinguishability of changing regions from the background. To enhance the sensitivity of the changing regions in the dual-temporal features, STSP-Net, proposed by He et al. [58], introduces a spatial attention fusion module. Wang et al. [59] use a difference enhancement module designed based on channel attention to enhance the difference features of different layers. Tang et al. [60] construct the ensemble channel attention module to fuse different levels of difference features, specifically by concatenating and adding difference features before feeding them into the channel attention module separately. Zhang et al. [48] present a dual correction attention module consisting of a parallel position-correlated attention module and a channel-correlated attention module to enhance the difference features at different stages. Zhang et al. [61] design a global attention module based on CBAM [62], which consists of cascaded channel attention and spatial attention to enhance the learning capability of contextual information of dual-temporal features. Liu et al. [63] introduce a spatial attention module to model the spatial context of the different layers of features generated by the Siamese backbone, and after multilayer feature fusion, they apply a channel attention module to model the channel context, thus enhancing the change region in space and channel, respectively.

C. Brain-Inspired Scheme for Remote Sensing Object-Level Change Detection

In the field of remote sensing image processing, in addition to the traditional convolutional neural network, the brain-inspired methods represented by spiking neural networks

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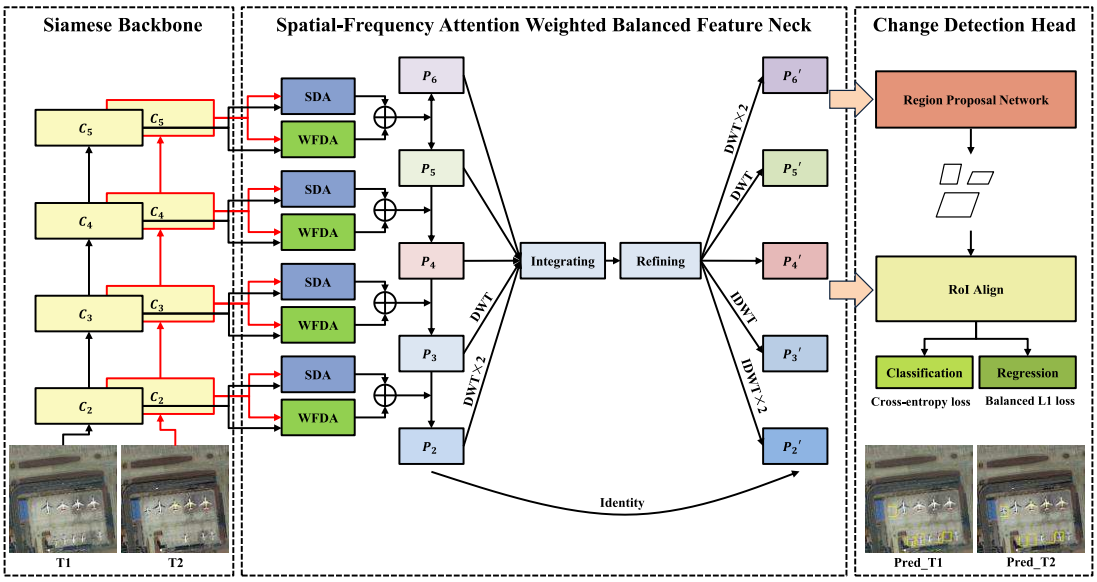


Fig. 1. Overall structure of our proposed SFDA-OCDet.

(SNNs) also show excellent ability. SNN is a neural network that mimics the impulse communication mechanism of biological neurons, utilizing the time and frequency of pulse issuance to encode information, featuring low power consumption and advantages in processing dynamic spatio-temporal information. Researchers have applied SNNs to remote sensing image analysis in a series of breakthroughs. Li et al. [64] design a spatio-temporal pruning training method to solve the problem of information loss in the case of ultralow latency of SNN, which significantly improves the performance of remote sensing image scene classification. Wu et al. [65] introduced SNN into remote sensing scene classification and innovatively proposed a multibit-based SNN, which effectively improved the classification accuracy. Liu et al. [66] propose an SNN-SSEM network incorporating a channel attention mechanism, and the high-precision and low-energy results achieve a bright performance in hyperspectral image classification datasets. Duan et al. [67] proposed an online adaptive scheme based on SNN for on-orbit satellite or airborne applications, which is able to adapt to changes in the environment and weather online, and achieved excellent capability in remote sensing image classification, segmentation, and detection. Yang et al. [68] propose for the first time a self-supervised information bottleneck learning method based on SNN, which is capable of realizing high-precision optical flow estimation in the presence of noise interference. These brain-inspired research efforts have provided us with reliable new ideas, and in our future work, we will deeply explore the application of SNN in remote sensing object-level change detection.

III. Methodology

A. Structure Overview

The structure of the object-level change detection network SFDA-OCDet proposed in this article is shown in Fig. 1. SFDA-OCDet consists of a Siamese backbone, spatial-frequency attention weighted balanced feature neck, and change detection head. In the Siamese backbone section, a pair of images at *T1* and *T2* phases is fed into Siamese ResNet50, sharing weight parameters to generate features at different stages of the dual-temporal images. In the neck section, SFDA-OCDet employs the SDA module and the WFDA module to enhance the difference features of each stage from the spatial and frequency domains, respectively, which are then fused and fed into the WBFP module for further refinement. In the change detection head section, SFDA-OCDet first generates a series of change proposals using a region proposal network (RPN), then exploits RoI Align to map the change proposals to a fixed size, and finally performs classification and regression to generate the category and location of the change objects.

B. SDA Module

The SDA module is proposed to enhance the learning ability of the network in the spatial domain with respect to the different features of the different stages produced by the Siamese backbone. The structure of the SDA module is shown in Fig. 2, and its design idea is inspired by AFF [69] and GCNet [70]. The inputs to the SDA module are the dual-temporal features generated by the Siamese backbone network at stages C_2, C_3, C_4, and C_5, respectively, denoted by $\mathbf{F}_1$ and $\mathbf{F}_2$, and the outputs are the difference features $\mathbf{F}_s$ enhanced in the spatial domain at each stage. The inputs and the output features have the same dimension, i.e., $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_s\} \in R^{C\times H\times W}$. The detailed operation of the SDA module is as follows.

First, the difference feature $\mathbf{F}_d$ of the dual-temporal feature is computed, which operates by subtracting the two features and calculating the absolute value, as shown in the following equation:

$$\mathbf{F}_d = |\mathbf{F}_1 \ominus \mathbf{F}_2| \tag{1}$$

where $|\bullet|$ denotes the computation of the absolute value operation and $\ominus$ denotes the element-wise subtraction operation.

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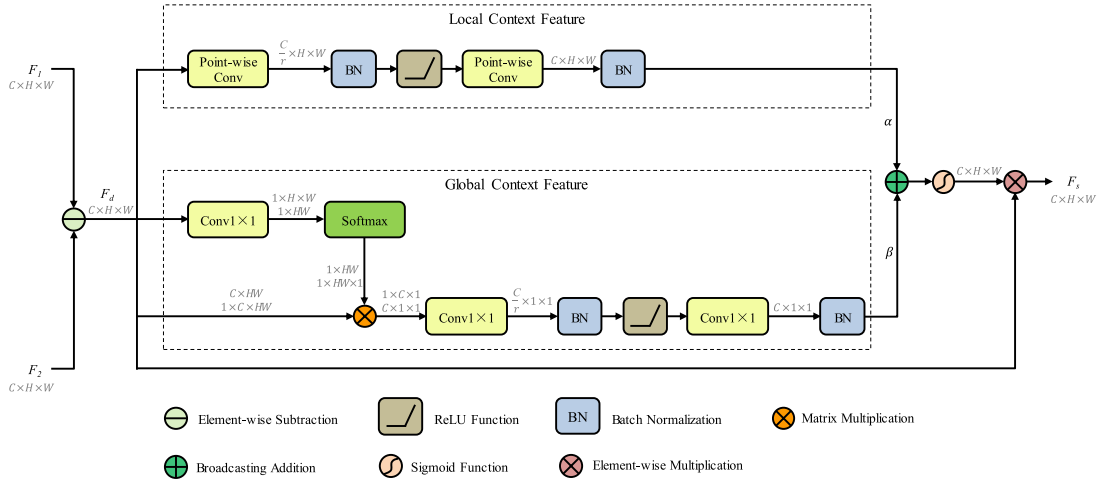


Fig. 2. Illustration of the structure of the SDA module.

Second, two parallel branches are utilized to generate the local context features and global context features of the difference features, respectively. Local feature branching introduces point-wise convolution (PWConv) to compute the interrelationships of each spatial location in the channel dimension to obtain the local context feature $\mathbf{L}(\mathbf{F}_d) \in \mathbb{R}^{C\times H\times W}$, which is formulated as shown below

$$\mathbf{L}(\mathbf{F}_d) = \mathrm{BN}\left(\mathrm{PWConv}_2(R(\mathrm{BN}(\mathrm{PWConv}_1(\mathbf{F}_d))))\right) \quad (2)$$

where the dimensions of the output feature maps of PWConv_1 and PWConv_2 are $(C/r)\times H\times W$ and $C\times H\times W$, respectively, BN denotes the batch normalization operation, and R denotes the ReLU function.

The global feature branch first uses 1×1 convolution with Softmax function to compute the attention weight, which is multiplied with the input difference feature to obtain the context information, and then, uses two 1×1 convolutions to perform the dimension transformation to obtain the global context feature $\mathbf{G}(\mathbf{F}_d) \in \mathbb{R}^{C\times 1\times 1}$. This calculation process can be summarized as follows:

$$\mathbf{G}_c = \mathrm{Softmax}\left(\mathrm{Conv}_{1\times 1}(\mathbf{F}_d)\right)\otimes \mathbf{F}_d \quad (3)$$

$$\mathbf{G}(\mathbf{F}_d) = \mathrm{BN}\left(\mathrm{Conv}^2_{1\times 1}\left(\mathrm{R}\left(\mathrm{BN}\left(\mathrm{Conv}^1_{1\times 1}(\mathbf{G}_c)\right)\right)\right)\right) \quad (4)$$

where $\mathrm{Conv}_{1\times 1}, \mathrm{Conv}^1_{1\times 1}$, and $\mathrm{Conv}^2_{1\times 1}$ represent the 1×1 convolution of the number of output channels 1, (C/r), and C, and $\otimes$ denotes the matrix multiplication operation. We followed the settings in AFF and GCNet, with r taking the value 4.

Finally, the local context features are fused with the global context features to obtain the attention weights in the spatial domain through the Sigmoid function, which are multiplied with the original difference feature to obtain the spatial domain-enhanced difference feature $\mathbf{F}_s$. To adaptively highlight the importance of local and global contextual features, we introduce two learnable weighting parameters α and β when fusing the two. The acquisition process of $\mathbf{F}_s$ is described as follows:

$$\mathbf{F}_s = \sigma\left(\alpha\bullet\mathbf{L}(\mathbf{F}_d)\oplus\beta\bullet\mathbf{G}(\mathbf{F}_d)\right)\otimes\mathbf{F}_d \quad (5)$$

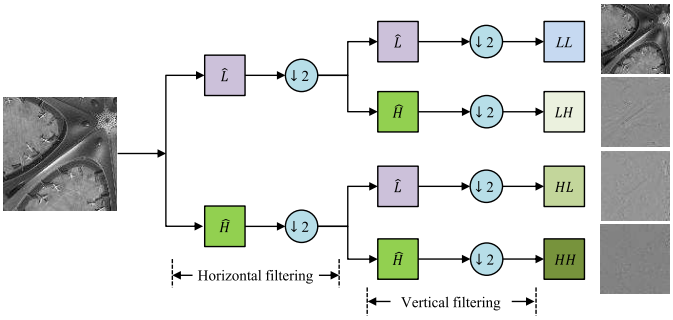


Fig. 3. Schematic of the DWT process.

where σ denotes the Sigmoid function, $\oplus$ represents broadcasting addition, and $\otimes$ is element-wise multiplication.

C. WFDA Module

The WFDA module is designed to enhance the network's ability to learn different features in the frequency domain using the DWT. DWT can decompose the difference features into low-frequency information and high-frequency information, which helps to refine the features of different frequencies. Moreover, DWT and inverse DWT (I) processes do not cause loss of information. Therefore, to take advantage of the DWT for frequency domain analysis, we propose the WFDA module by combining the DWT with the attention mechanism.

The process of DWT of a 2-D image is shown in Fig. 3. The input to the DWT is a 2-D image or feature, and the output is a low frequency component LL and three high frequency components LH, HL, HH. LL denotes the approximation of the original image, and LH, HL, and HH contain the detailed information of horizontal edges, vertical edges, and diagonal edges, respectively. The process of DWT is to first input each row of data in the horizontal direction to the low-pass filter $\hat{L}$ and the high − pass filter $\hat{H}$, respectively, and sample them in the ratio of 2:1 to obtain the preliminary result of the first half of the horizontal direction being the low-frequency component and the second half being the high-frequency component. The preliminary results are then passed through a low-pass filter

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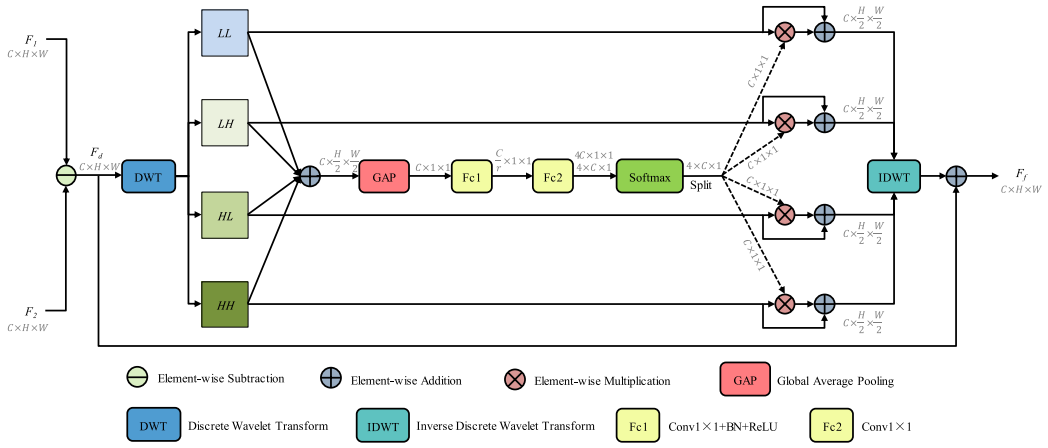


Fig. 4. Illustration of the structure of the WFDA module.

and a high-pass filter in the vertical direction for each column of data, resulting in four low-frequency and high-frequency components of halved size.

We use the Haar wavelet to implement DWT because the Haar wavelet has the advantages of simplicity and efficiency, orthogonality, and so on, which can make the arithmetic process reversible. In the Haar wavelet transform, the representation matrices of the low-pass filter f_{LL} and high − pass filters f_{LH}, f_{HL}, f_{HH} are as follows:

$$f_{LL} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad f_{\mathrm{LH}} = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \tag{6}$$

$$f_{\mathrm{HL}} = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}, \quad f_{\mathrm{HH}} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \tag{7}$$

Assuming that $\mathbf{x}$ denotes the input image, the output of $\mathbf{x}$ after the Haar wavelet transform and the computational process can be described using the following equation:

$$\begin{cases} \mathrm{DWT}(\mathbf{x}) = \mathbf{x}_{LL}, \mathbf{x}_{\mathrm{LH}}, \mathbf{x}_{\mathrm{HL}}, \mathbf{x}_{\mathrm{HH}} \\ \mathbf{x}_{LL} = \left(\mathbf{x} \circledast f_{LL}\right) \downarrow 2 \\ \mathbf{x}_{\mathrm{LH}} = \left(\mathbf{x} \circledast f_{\mathrm{LH}}\right) \downarrow 2 \\ \mathbf{x}_{\mathrm{HL}} = \left(\mathbf{x} \circledast f_{\mathrm{HL}}\right) \downarrow 2 \\ \mathbf{x}_{\mathrm{HH}} = \left(\mathbf{x} \circledast f_{\mathrm{HH}}\right) \downarrow 2 \end{cases} \tag{8}$$

where $\circledast$ denotes a convolution operation and $\downarrow 2$ denotes a standard downsampling operation with a factor of 2.

The value of the low-frequency component and the high-frequency component of the Haar wavelet transform output at any (i, j) position can be calculated using the following equations:

$$\begin{aligned} \mathbf{x}_{LL}(i,j) = {} & \mathbf{x}(2i{-}1, 2j-1) + \mathbf{x}(2i{-}1, 2j) \\ & + \mathbf{x}(2i, 2j-1) + \mathbf{x}(2i, 2j) \end{aligned} \tag{9}$$

$$\begin{aligned} \mathbf{x}_{\mathrm{LH}}(i,j) = {} & -\mathbf{x}(2i{-}1, 2j-1) - \mathbf{x}(2i{-}1, 2j) \\ & + \mathbf{x}(2i, 2j-1) + \mathbf{x}(2i, 2j) \end{aligned} \tag{10}$$

$$\begin{aligned} \mathbf{x}_{\mathrm{HL}}(i,j) = {} & -\mathbf{x}(2i{-}1, 2j-1) + \mathbf{x}(2i{-}1, 2j) \\ & - \mathbf{x}(2i, 2j-1) + \mathbf{x}(2i, 2j) \end{aligned} \tag{11}$$

$$\begin{aligned} \mathbf{x}_{\mathrm{HH}}(i,j) = {} & \mathbf{x}(2i{-}1, 2j-1) - \mathbf{x}(2i{-}1, 2j) \\ & - \mathbf{x}(2i, 2j-1) + \mathbf{x}(2i, 2j). \end{aligned} \tag{12}$$

The process of IDWT is opposite to DWT, and in accordance with the reversibility of the Haar wavelet transform, it can be reconstructed as the original image according to the following equations:

$$\mathbf{x}(2i{-}1, 2j-1) = \begin{bmatrix} \mathbf{x}_{LL}(i,j) - \mathbf{x}_{\mathrm{LH}}(i,j) \\ -\mathbf{x}_{\mathrm{HL}}(i,j) + \mathbf{x}_{\mathrm{HH}}(i,j) \end{bmatrix} /\mathbf{4} \tag{13}$$

$$\mathbf{x}(2i{-}1, 2j) = \begin{bmatrix} \mathbf{x}_{LL}(i,j) - \mathbf{x}_{\mathrm{LH}}(i,j) \\ +\mathbf{x}_{\mathrm{HL}}(i,j) - \mathbf{x}_{\mathrm{HH}}(i,j) \end{bmatrix} /4 \tag{14}$$

$$\mathbf{x}(2i, 2j-1) = \begin{bmatrix} \mathbf{x}_{LL}(i,j) + \mathbf{x}_{\mathrm{LH}}(i,j) \\ -\mathbf{x}_{\mathrm{HL}}(i,j) - \mathbf{x}_{\mathrm{HH}}(i,j) \end{bmatrix} /4 \tag{15}$$

$$\mathbf{x}(2i, 2j) = \begin{bmatrix} \mathbf{x}_{LL}(i,j) + \mathbf{x}_{\mathrm{LH}}(i,j) \\ +\mathbf{x}_{\mathrm{HL}}(i,j) + \mathbf{x}_{\mathrm{HH}}(i,j) \end{bmatrix} /4. \tag{16}$$

We incorporate the DWT and IDWT into the object-level change detection network and design the WFDA module as shown in Fig. 4 by combining the attention mechanism. The input to the WFDA module is the dual-temporal features produced by the Siamese backbone in each of the four stages, and the output is the difference feature $\mathbf{F}_f$ in the frequency domain after enhancement of each stage. The dimensions of the input and output features remain the same, i.e., $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_f\} \in \mathbb{R}^{C\times H\times W}$. The computational process of the WFDA module can be described as follows.

First, the difference feature $\mathbf{F}_d$ of the dual-temporal feature is calculated, and a DWT operation is performed on $\mathbf{F}_d$ to obtain the low-frequency component $\mathbf{F}_d^{LL}$ and the high-frequency components $\mathbf{F}_d^{\mathrm{LH}}, \mathbf{F}_d^{\mathrm{HL}}$ and $\mathbf{F}_d^{\mathrm{HH}}$ of the difference feature. $\mathbf{F}_d^{LL}, \mathbf{F}_d^{\mathrm{LH}}, \mathbf{F}_d^{\mathrm{HL}}, \mathbf{F}_d^{\mathrm{HH}} \in R^{C\times(H/2)\times(W/2)}$

$$\mathbf{F}_d^{LL}, \mathbf{F}_d^{\mathrm{LH}}, \mathbf{F}_d^{\mathrm{HL}}, \mathbf{F}_d^{\mathrm{HH}} = \mathrm{DWT}(\mathbf{F}_d). \tag{17}$$

Second, the low-frequency components of the difference features are fused with the high-frequency components, and the fused features are passed through global average pooling, two fully connected layers, a Softmax function, and the split operation to obtain the attention weights of each component. The details of the process are as follows:

$$\mathbf{F}_d^U = \mathbf{F}_d^{LL} + \mathbf{F}_d^{\mathrm{LH}} + \mathbf{F}_d^{\mathrm{HL}} + \mathbf{F}_d^{\mathrm{HH}} \tag{18}$$

$$W_d^U = \mathrm{Softmax}\left(Fc_2\left(Fc_1\left(\mathrm{GAP}\left(\mathbf{F}_d^U\right)\right)\right)\right) \tag{19}$$

$$W_d^{LL}, W_d^{\mathrm{LH}}, W_d^{\mathrm{HL}}, W_d^{\mathrm{HH}} = \mathrm{split}\left(W_d^U\right) \tag{20}$$

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where GAP stands for global average pooling, Fc_1 consists of Conv1 $\times$ 1, BN layer and ReLU function, whose role is to downscale the number of channels from C to $(C/r), r$ takes the value of 4, Fc_2 represents Conv1 $\times$ 1, which is used to increase the number of channels to $4C$, split operation is used to split the weights output from the Softmax function into four component weights.

Finally, the attention weights of each component are multiplied and then summed with the corresponding low-frequency and high-frequency components. The IDWT operation is performed on the weighted low-frequency and high-frequency components, and then the transform results are residually connected with the initial difference feature $\mathbf{F}_d$ to obtain the difference feature $\mathbf{F}_f$ after enhancement in the frequency domain. The process of the above operation can be summarized by the following equations:

$$\mathbf{F}_d^{'i} = W_d^i \otimes \mathbf{F}_d^i + \mathbf{F}_d^i, i = LL, \mathrm{LH}, \mathrm{HL}, \mathrm{HH} \tag{21}$$

$$\mathbf{F}_f = \mathrm{IDWT}\left(\mathbf{F}_d^{'LL}, \mathbf{F}_d^{'\mathrm{LH}}, \mathbf{F}_d^{'\mathrm{HL}}, \mathbf{F}_d^{'\mathrm{HH}}\right) + \mathbf{F}_d. \tag{22}$$

D. Wavelet BFP

The balanced feature pyramid (BFP) proposed by Libra R-CNN [71] uses max pooling and interpolation for downsampling and upsampling operations several times in the process of refining multilevel features, which will result in the loss of feature information and negatively affect the change detection accuracy. DWT and IDWT can also implement the downsampling and upsampling process of the features, and the reversibility of the wavelet transform makes the decomposition and reconstruction process of the features without losing the feature information. Therefore, we introduce DWT and IDWT into BFP and propose the WBFP module to solve the problem of information loss during multilevel feature refinement processing and further improve the change detection accuracy. As shown in Fig. 1, WBFP consists of four parts: rescaling, integrating, refining, and strengthening.

The input features of WBFP are the five features $P_2, P_3, P_4, P_5,$ andP_6 generated by the FPN neck, and the output features are the refined features $P_2', P_3', P_4', P_5',$ andP_6'. $P_2 \sim P_6$ have the same number of channels, and the feature size is gradually halved.

The rescaling process is to resize the $P_2 \sim P_6$ features uniformly to the size of the intermediate feature P_4. Specifically, two DWT processes are performed for P_2, one DWT process is performed for P_3, and the same interpolation operation as BFP is used for P_5 and P_6. The corresponding process is represented by the following equation:

$$\begin{cases} P_2^{LL_1}, P_2^{\mathrm{LH}_1}, P_2^{\mathrm{HL}_1}, P_2^{\mathrm{HH}_1} = \mathrm{DWT}\left(P_2\right) \\ P_2^{LL_2}, P_2^{\mathrm{LH}_2}, P_2^{\mathrm{HL}_2}, P_2^{\mathrm{HH}_2} = \mathrm{DWT}\left(P_2^{LL_1}\right) \\ P_3^{LL}, P_3^{\mathrm{LH}}, P_3^{\mathrm{HL}}, P_3^{\mathrm{HH}} = \mathrm{DWT}\left(P_3\right) \\ P_4 = P_4 \\ P_5 = \mathrm{Interpolate}\left(P_5\right) \\ P_6 = \mathrm{Interpolate}\left(P_6\right) \end{cases} \tag{23}$$

The integrating process is the averaging of multilevel features unified to the same size to obtain a balanced semantic feature (BSF)

$$\mathrm{Bsf} = \left(P_2^{LL_2} + P_3^{LL} + P_4 + P_5 + P_6\right)/5. \tag{24}$$

The refining process is a refinement of the BSFs using the nonlocal module.

The strengthening process involves restoring the refined BSF to the size of the original feature and performing a residual connection with the original feature, thus completing the refinement of the multilevel features. Specifically, two IDWTs are performed on the BSF to recover to size P_2, one IDWT to recover to size P_3, one DWT to recover to size P_5, and two DWTs to recover to size P_6. This process can be described by the following equations:

$$\begin{cases} P_2' = \mathrm{IDWT}\left(\mathrm{Bsf}, P_2^{\mathrm{LH}_2}, P_2^{\mathrm{HL}_2}, P_2^{\mathrm{HH}_2}\right) \\ P_2' = \mathrm{IDWT}\left(P_2', P_2^{\mathrm{LH}_1}, P_2^{\mathrm{HL}_1}, P_2^{\mathrm{HH}_1}\right) + P_2 \end{cases} \tag{25}$$

$$P_3' = \mathrm{IDWT}\left(\mathrm{Bsf}, P_3^{\mathrm{LH}}, P_3^{\mathrm{HL}}, P_3^{\mathrm{HH}}\right) + P_3 \tag{26}$$

$$P_4' = \mathrm{Bsf} + P_4 \tag{27}$$

$$P_5' = P_5^{LL} + P_5 \tag{28}$$

$$P_6' = P_6^{LL_2} + P_6 \tag{29}$$

where

$$\begin{cases} P_5^{LL}, P_5^{\mathrm{LH}}, P_5^{\mathrm{HL}}, P_5^{\mathrm{HH}} = \mathrm{DWT}\left(\mathrm{Bsf}\right) \\ P_6^{LL_1}, P_6^{\mathrm{LH}_1}, P_6^{\mathrm{HL}_1}, P_6^{\mathrm{HH}_1} = \mathrm{DWT}\left(\mathrm{Bsf}\right) \\ P_6^{LL_2}, P_6^{\mathrm{LH}_2}, P_6^{\mathrm{HL}_2}, P_6^{\mathrm{HH}_2} = \mathrm{DWT}\left(P_6^{LL_1}\right). \end{cases} \tag{30}$$

E. Loss Function

The loss function used in this article is consistent with Libra R-CNN and consists of classification loss and regression loss. Among them, the classification loss function is cross-entropy loss, and the regression loss function is balanced $L1$ loss. Balanced $L1$ loss [71] can facilitate critical regression gradients that can rebalance the samples in the dataset as well as the joint training process for classification and regression tasks, which can be described by the following equation:

$$balancedL_1\left(x\right) = \begin{cases} \dfrac{\alpha}{b}\left(b|x|+1\right)\ln\left(b|x|+1\right) - \alpha|x|, & if\,|x| < 1 \\ \gamma|x| + C, & \text{otherwise} \end{cases} \tag{31}$$

where the parameters α, b, γ satisfy $\alpha \ln\left(b+1\right) = \gamma$, the values of α and γ are 0.5 and 1.5, respectively, C denotes a constant.

IV. Experiments and Analysis

A. Dataset Description

To evaluate our proposed SFDA-OCDet, we construct four object-level change detection datasets, Airplane-CD, Vehicle-CD, Building-CD, and Wind turbine-CD, using four typical targets, airplanes, vehicles, buildings, and wind turbines, respectively.

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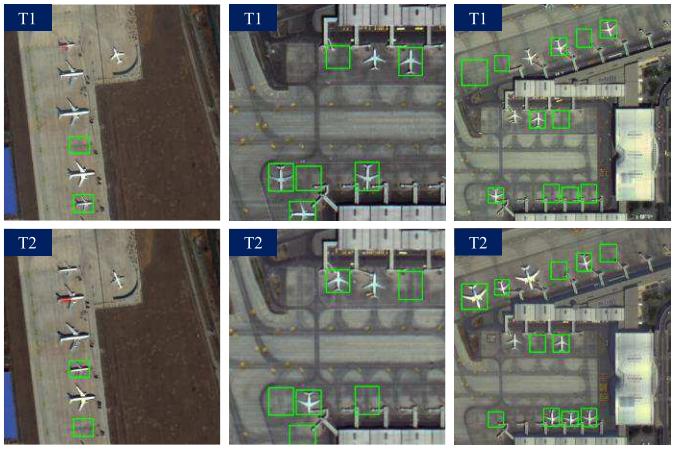


Fig. 5. Examples of image pairs in the Airplane-CD dataset. (*T1*) Image at *T1* time-phase. (*T2*) Image at *T2* time-phase.

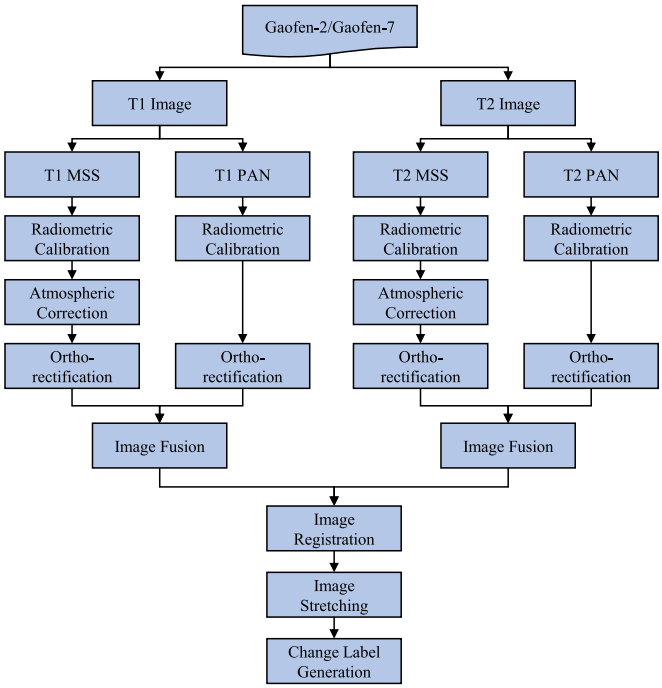


Fig. 6. Pre-processing flow for Gaofen-2 and Gaofen-7 data.

1) Airplane-Cd Dataset: The images in the Airplane-CD dataset are collected from 60 major airports in China, including Shanghai Pudong International Airport, Beijing Daxing International Airport, Beijing Capital International Airport, and so on. The dataset consists of 134 pairs of original-size dual-temporal remote sensing images from both Gaofen-2 and Gaofen-7 data sources, including 83 pairs of Gaofen-2 images and 51 pairs of Gaofen-7 images. The resolution of Gaofen-2 and Gaofen-7 images is 1 and 0.65 m, respectively. We seamlessly crop all the original size images to obtain a total of 1150 pairs of images of size 512×512, which are then randomly divided into training, validation, and test sets in the ratio of 5:2:3, containing 575, 230, and 345 pairs of images, respectively. Fig. 5 shows some pairs of dual-temporal images labeled with changed airplane objects.

For the data pre-processing, we use ENVI 5.6 to complete the pre-processing process of Gaofen-2 and Gaofen-7 data, including radiometric calibration, atmospheric correction, ortho-rectification, image fusion, image registration, and image stretching. Radiometric calibration is used to convert digital numbers recorded by remote sensors to radiance. The purpose of atmospheric correction is to eliminate the effects produced by factors such as atmosphere and light. The role of ortho-rectification is to generate ortho-rectified images by using ground control points (GCPs) and digital elevation models (DEMs) to eliminate geometric distortions produced by factors such as terrain relief and remote sensor attitude. Image fusion is performed using the NNDiffuse pan sharpening algorithm to obtain fused images with both high resolution and multispectral characteristics. We complete the registration of the different phase images using the Image Registration Workflow tool of ENVI 5.6 to achieve geometric alignment between these images. We use the optimized linear image stretching algorithm of ENVI 5.6 to convert 16-bit images to 8-bit images. Fig. 6 shows our detailed pre-processing flow for Gaofen-2 and Gaofen-7 data. First, radiometric calibration, atmospheric correction, and ortho-rectification are performed on the multispectral (MSS) image of *T1* time-phase; radiometric calibration and ortho-rectification are performed on the panchromatic (PAN) image of *T1* time-phase, and the fused image of *T1* time-phase is obtained after image fusion; then, the same process is performed on the image of *T2* time-phase, and the fused image of *T2* time-phase is obtained; finally, the fused images of dual time-phases undergo image registration and image stretching. After completing the above pre-processing, the label of the changed aircraft is obtained through the change label generation process.

Fig. 7 displays the schematic of the change label generation process. The core idea is to determine the position of the changed airplanes by comparing the rectangular boxes of the airplanes in *T1* image with the rectangular boxes of the airplanes in *T2* image, and the detailed algorithm is shown in Table I. As shown in Fig. 7, the rectangular boxes of the changed airplanes can be obtained by the change label generation algorithm as {B, C, E, F, G, I, K}.

We use the algorithm described in Table I to generate labels for the changed objects of the Airplane-CD, Vehicle-CD, and Wind turbine-CD datasets. For the Building-CD dataset, we generate labels using pixel-level masks of changed buildings.

2) Vehicle-Cd Dataset: We create the Vehicle-CD dataset by manually labeling the changing vehicle targets on the dual-temporal images of the building change detection dataset WHU-CD [72]. The Vehicle-CD dataset contains a total of 668 pairs of images of size 512×512 with a resolution of 0.2 m. The training set contains 334 pairs of images, the validation set contains 134 pairs of images, and the test set contains 200 pairs of images, based on a random division in the ratio of 5:2:3. Fig. 8 displays some pairs of dual-temporal images labeled with changed vehicle targets.

3) Building-Cd Dataset: We construct the object-level building change detection dataset Building-CD using the pixel-level building change detection dataset LEVIR-CD [73]. The

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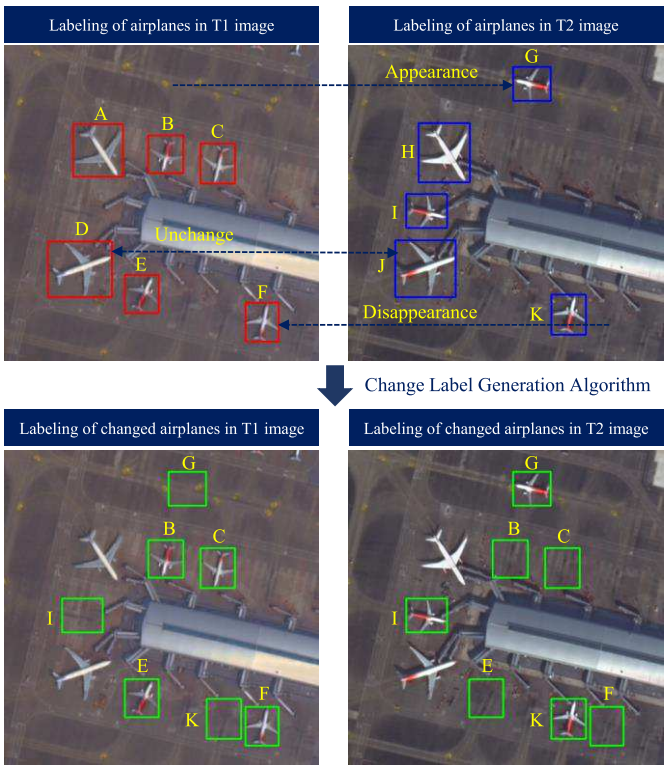


Fig. 7. Schematic of the change label generation process.



Fig. 8. Examples of image pairs in the Vehicle-CD dataset. (*T1*) Image at *T1* time-phase. (*T2*) Image at *T2* time-phase.

original LEVIR-CD dataset contains 637 Google Earth images of size 1024 × 1024 with a resolution of 0.5 m. Its training, validation, and test sets contain 445, 64, and 128 image pairs, respectively. Based on the original data division, we seamlessly crop the data according to the size of 512×512, and finally obtained 1168 pairs of training set images, 168 pairs of validation set images, and 334 pairs of test set images. We generate horizontal minimum outer rectangles for the pixel-level changed ground truth of the building, thus obtaining the corresponding object-level changed ground truth. Fig. 9 provides some pairs of dual-temporal images labeled with changed building targets.

TABLE I

DETAILED ALGORITHM FOR GENERATING CHANGED RECTANGULAR BOX LABELS

Algorithm Change Label Generation

Input: $M = \{m_i\}$ is the list of rectangular boxes of the airplanes in T1 image
$N = \{n_j\}$ is the list of rectangular boxes of the airplanes in T2 image

Output: $O = \{o_k\}$ is the list of rectangular boxes of the changed airplanes between T1 and T2 images.

O is an empty list;
Unchanged_M is an empty list used for saving rectangular boxes of the unchanged airplanes in M;
Unchanged_N is an empty list used for saving rectangular boxes of the unchanged airplanes in N;
thresh = 0.5;
for *m in M* **do**
  for *n in N* **do**
    iou = *ComputeIoU(m, n)*
    if *iou* > *thresh* **then**
      Unchanged_M.append(*m*)
      Unchanged_N.append(*n*)
    end if
  end for
end for
if *Unchanged_M is not empty* **then**
  for *u_m in Unchanged_M* **do**
    delete u_m from M
  end for
  for *u_n in Unchanged_N* **do**
    delete u_n from N
  end for
end if
$O = M + N$
return O

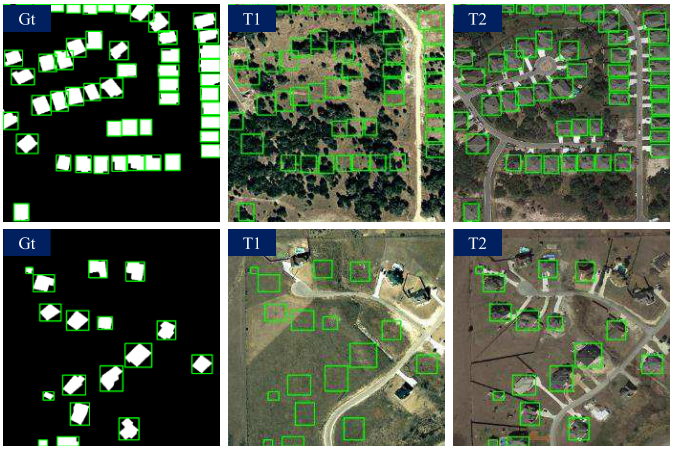


Fig. 9. Examples of image pairs in the Building-CD dataset. (Gt) Pixel-level changed ground truth and its minimum outer rectangles. (*T1*) Image at *T1* time-phase. (*T2*) Image at *T2* time-phase.

4) Wind Turbine-Cd Dataset: We collect satellite images of different time phases from seven provinces where wind farms are mainly distributed, including Xinjiang, Gansu, Qinghai,

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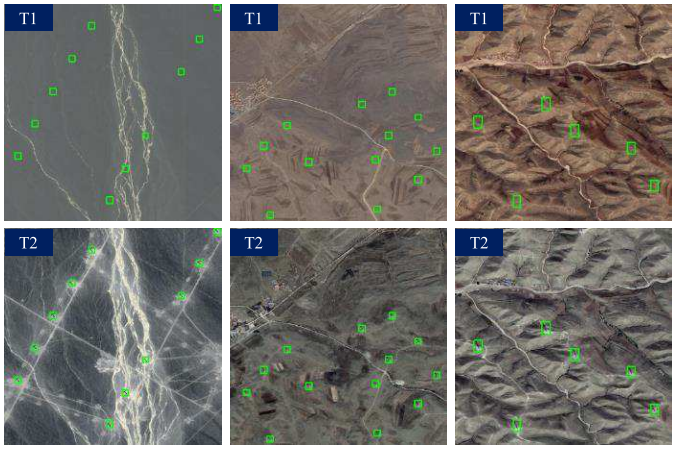


Fig. 10. Examples of image pairs in the Wind turbine-CD dataset. (*T1*) Image at *T1* time-phase. (*T2*) Image at *T2* time-phase.

Ningxia, Inner Mongolia, Hebei, and Shanxi, which cover complex backgrounds such as the desert, grasslands, and mountains. The satellite data source is Google Earth, and the resolution is between 3 and 4 m. We acquire 191 pairs of dual-temporal images with original size distribution between 1280×1792 and 8704×4608. In remote sensing images, wind turbines and their shadows together constitute typical interpretation features; therefore, consistent with the research [74], [75], [76], we annotate the wind turbine together with its shadow in the rectangular box. We crop all the original size images to 10488 pairs of images with a size of 512×512 based on an overlapping area of 256. The training set, validation set, and test set obtained by dividing the data in a ratio of 5:2:3 contain 5244, 2098, and 3146 pairs of images, respectively. Fig. 10 gives some pairs of dual-temporal images labeled with changed wind turbines.

B. Experiment Settings

Our proposed SFDA-OCDet and other comparison methods are implemented based on PyTorch [77] and MMDetection [78]. All experiments are finished on NVIDIA GeForce RTX 3060 GPUs (12G) and the same experimental settings are adopted to ensure a fair comparison. The optimizer used for the experiment is AdamW with a weight decay of 0.0001. The batch size is 2 pairs of dual-temporal images. The learning rate has an initial value of 0.0002, which decreases by a factor of 10 at the 24th epoch and 33rd epoch for Airplane-CD, Vehicle-CD, and Building-CD datasets, and the 8th and 11th epoch for Wind turbine-CD dataset. We train all methods on Airplane-CD, Vehicle-CD, Building-CD, and Wind turbine-CD datasets for 36, 36, 36, and 12 epochs, respectively. Only a random flip has been employed for data enhancement of dual-temporal images.

C. Evaluation Metrics

Due to the differences between object-level change detection and pixel-level change detection in the representation of change regions, the evaluation metrics for pixel-level change detection are no longer applicable, and it is more appropriate to use the evaluation methods used in the object detection task. Therefore, we utilize average precision (AP) used by MS COCO [79] as an evaluation metric for each object-level change detection method.

The formula for AP is shown below

$$\mathrm{AP} = \int_0^1 P(R)\, dR \tag{32}$$

where P denotes precision and R represents recall, P and R can be calculated using the following equations:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{33}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{34}$$

where true positive (TP) indicates the number of bounding boxes that are detected as changed targets and are detected correctly, false positive (FP) indicates the number of bounding boxes that are detected as changed targets but are detected incorrectly, and false negative (FN) indicates the number of changed targets that are missed.

We chose the more refined metrics $\mathrm{AP}_{0.5}$ and $\mathrm{AP}_{0.5:0.95}$ to evaluate the performance of each method. $\mathrm{AP}_{0.5}$ indicates the AP value when the IoU of the change detection bounding box and the ground truth is 0.5. $\mathrm{AP}_{0.5:0.95}$ denotes the average value of each AP when IoU is incremented from 0.5 to 0.95 in steps of 0.05.

D. Ablation Analysis

To validate the effectiveness of the modules in the proposed SFDA-OCDet, we conduct ablation experiments on the Airplane-CD dataset, and the specific experimental results and analysis are described in this section.

1) *Baseline*: We transform the object detection model Libra R-CNN into an object-level change detection network, name it Libra R-CNN_cd, and use it as the baseline model for the experimental analysis in this article. Libra R-CNN_cd chooses the Siamese tResNet50 as the backbone and uses the commonly used subtraction and absolute value to compute the difference features, and then utilizes the FPN as the neck network, whose detection head is consistent with Libra R-CNN. The baseline model achieves 96.81% and 77.60% for $\mathrm{AP}_{0.5}$ and $\mathrm{AP}_{0.5:0.95}$ on the Airplane-CD dataset, respectively.
2) *Effect of SDA:* The commonly used way of obtaining difference features by subtracting and calculating the absolute value is simple and has limited ability to represent the difference features of changed objects in a dual-temporal image. To solve this problem, we design the SDA module in SFDA-OCDet with the aim of enhancing the difference features in the spatial domain and improving the object-level change detection accuracy. To validate the effectiveness of the SDA module, we conduct ablation experiments on the Airplane-CD dataset. The results in the second row of Table II show that, after adding the SDA module to the baseline model, $\mathrm{AP}_{0.5}$ is increased from 96.81% to 97.72%, and $\mathrm{AP}_{0.5:0.95}$

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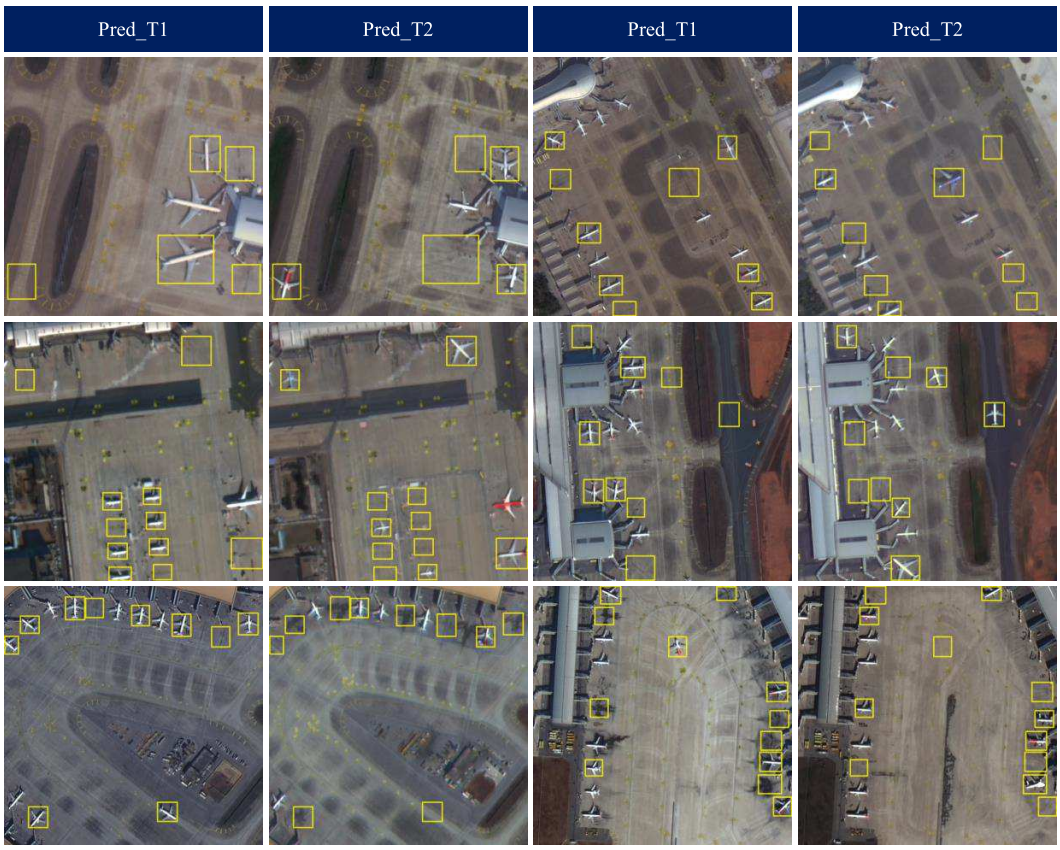


Fig. 11. Example of visualized detection results of the proposed SFDA-OCDet on the Airplane-CD dataset. (Pred_*T1*) and (Pred_*T2*) change detection results shown on *T1* and *T2* time-phase images.

TABLE II
ABLATION EXPERIMENTS ON THE AIRPLANE-CD DATASET

Baseline	SDA	WFDA	WBFP	$AP_{0.5}$(%)	$AP_{0.5:0.95}$(%)	Params(M)	FLOPs(G)	Inference Time (s)
✓				96.81	77.60	41.39	72.15	0.0571
✓	✓			97.72	77.87	46.99	74.34	0.0600
✓		✓		97.75	78.02	48.37	72.16	0.0652
✓			✓	97.75	77.75	41.39	72.15	0.0647
✓	✓	✓		98.50	78.16	53.97	74.35	0.0683
✓	✓		✓	97.77	78.13	46.99	74.34	0.0682
✓		✓	✓	97.80	78.03	48.37	72.16	0.0737
✓	✓	✓	✓	**98.60**	**78.33**	53.97	74.35	0.0786

is increased from 77.60% to 77.87%, which proves that the SDA module is able to enhance the different features from the spatial domain to effectively improve the change detection accuracy of the airplane.

3) *Effect of WFDA:* To further enhance the expression of different features, we propose a WFDA module for enhancing different features from the frequency domain using DWT and add it to the SFDA-OCDet model. Similarly, the validity is verified by comparing the change in accuracy before and after adding the WFDA module to the baseline model. The third row of Table II shows that after the introduction of the WFDA module, $AP_{0.5}$ and $AP_{0.5:0.95}$ are improved by 0.94% and 0.42%, respectively, which indicates the effectiveness of the WFDA module.

4) *Effect of WBFP:* The BFP repeatedly employs downsampling and upsampling operations with larger multiples, which can cause the problem of feature information loss. For this reason, we introduce DWT and IDWT to design the WBFP module. The fourth row of Table II shows the experimental results of the baseline model after adding the WBFP module, $AP_{0.5}$ is improved from 96.81% to 97.75%, and $AP_{0.5:0.95}$ is improved from 77.60% to 77.75%, which indicates that the WBFP module is effective in reducing the loss of information and improving the accuracy of change detection.

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Fig. 12. Example of visualized detection results of the proposed SFDA-OCDet on the Vehicle-CD dataset. (Pred_*T1*) and (Pred_*T2*) change detection results shown on *T1* and *T2* time-phase images.

5) *Effect of Module Combination:* The fifth to last row of Table II shows the results of the ablation experiments with different combinations of the three modules, SDA, WFDA, and WBFP, based on the baseline model. The accuracy of adding any two modules is superior to the accuracy of using either module alone. In the two-by-two combination, the best effect is achieved when the SDA module and WFDA module are used together, with $AP_{0.5}$ and $AP_{0.5:0.95}$ reaching 98.50% and 78.16%, respectively, which indicates that the joint enhancement of difference features in both spatial and frequency domains can achieve better results. When all three modules are introduced on the basis of baseline model, the complete SFDA-OCDet model is formed, and the accuracy reaches the highest, which is 98.60% and 78.33% for $AP_{0.5}$ and $AP_{0.5:0.95}$, respectively, which is 1.79% and 0.73% higher than the baseline model, respectively.

E. Comparison Analysis

To compare the performance of SFDA-OCDet with other methods, we conduct comparative experiments on the four datasets. We adapt 12 mainstream object detection models for object-level change detection, including SSD [80], RetinaNet [81], YOLOv3 [82], FCOS [83], Yolact [84], QueryInst [85], six one-stage models, and Faster R-CNN [86], Mask R-CNN [87], Libra R-CNN [71], TridentNet [88], Dynamic R-CNN [89], and Sparse R-CNN [90], six two-stage models. So as to distinguish from the names of the original object detection models, we add the suffix "_cd" to the original names (e.g., Faster R-CNN_cd and RetinaNet_cd). For the backbone setup, Siamese ResNet50 is used for all methods, except for SSD_cd and YOLOv3_cd, which use Siamese VGG16 and Siamese DarkNet53, respectively.

1) *Results on Airplane-CD Dataset:* The comparison results of our proposed SFDA-OCDet with other methods on the Airplane-CD dataset are shown in Table III. In the airplane change detection task, SFDA-OCDet achieved 98.60% for $AP_{0.5}$ and 78.33% for $AP_{0.5:0.95}$, which is the best performance among all the compared methods, and verified the effectiveness of the method. Fig. 11 shows the visualized detection results of SFDA-OCDet on the Airplane-CD dataset. The results show that from pre-temporal to post-temporal images, SFDA-OCDet can accurately locate the position of the changed airplane regardless of whether the airplane is appearing or disappearing, and can effectively detect the changes of airplanes with different sizes.
2) *Results on Vehicle-CD Dataset:* Table IV shows the test results of SFDA-OCDet with other methods on the Vehicle-CD dataset. SFDA-OCDet achieves 94.32%

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Fig. 13. Example of visualized detection results of the proposed SFDA-OCDet on the Building-CD dataset. (Pred_*T1*) and (Pred_*T2*) change detection results shown on *T1* and *T2* time-phase images.

TABLE III
COMPARISON OF THE ACCURACY OF EACH MODEL ON THE AIRPLANE-CD DATASET

Methods	Backbone	$AP_{0.5}$(%)	$AP_{0.5:0.95}$(%)
One-stage:			
SSD_cd	VGG16*	97.59	73.80
RetinaNet_cd	ResNet50*	97.81	78.10
YOLOv3_cd	DarkNet53*	95.16	67.17
FCOS_cd	ResNet50*	97.49	75.43
Yolact_cd	ResNet50*	96.50	69.45
QueryInst_cd	ResNet50*	88.90	69.04
Two-stage:			
Faster R-CNN_cd	ResNet50*	97.84	77.57
Mask R-CNN_cd	ResNet50*	97.82	77.89
Libra R-CNN_cd	ResNet50*	96.81	77.60
TridentNet_cd	ResNet50*	96.70	74.64
Dynamic R-CNN_cd	ResNet50*	96.74	78.13
Sparse R-CNN_cd	ResNet50*	88.50	68.73
SFDA-OCDet (Ours)	ResNet50*	**98.60**	**78.33**

* indicates the adoption of a Siamese structure for the backbone network.

TABLE IV
COMPARISON OF THE ACCURACY OF EACH MODEL ON THE VEHICLE-CD DATASET

Methods	Backbone	$AP_{0.5}$(%)	$AP_{0.5:0.95}$(%)
One-stage:			
SSD_cd	VGG16*	93.51	59.65
RetinaNet_cd	ResNet50*	91.25	55.37
YOLOv3_cd	DarkNet53*	92.01	55.10
FCOS_cd	ResNet50*	91.72	55.93
Yolact_cd	ResNet50*	92.36	55.51
QueryInst_cd	ResNet50*	81.60	51.35
Two-stage:			
Faster R-CNN_cd	ResNet50*	94.19	60.40
Mask R-CNN_cd	ResNet50*	93.14	60.21
Libra R-CNN_cd	ResNet50*	93.74	61.02
TridentNet_cd	ResNet50*	87.78	51.41
Dynamic R-CNN_cd	ResNet50*	91.97	60.79
Sparse R-CNN_cd	ResNet50*	82.13	52.01
SFDA-OCDet (Ours)	ResNet50*	**94.32**	**62.28**

* indicates the adoption of a Siamese structure for the backbone network.

and 62.28% in the metrics of $AP_{0.5}$ and $AP_{0.5:0.95}$, respectively, and obtains the best performance among the compared methods. In the Vehicle-CD dataset, the vehicles are small in size and belong to the category

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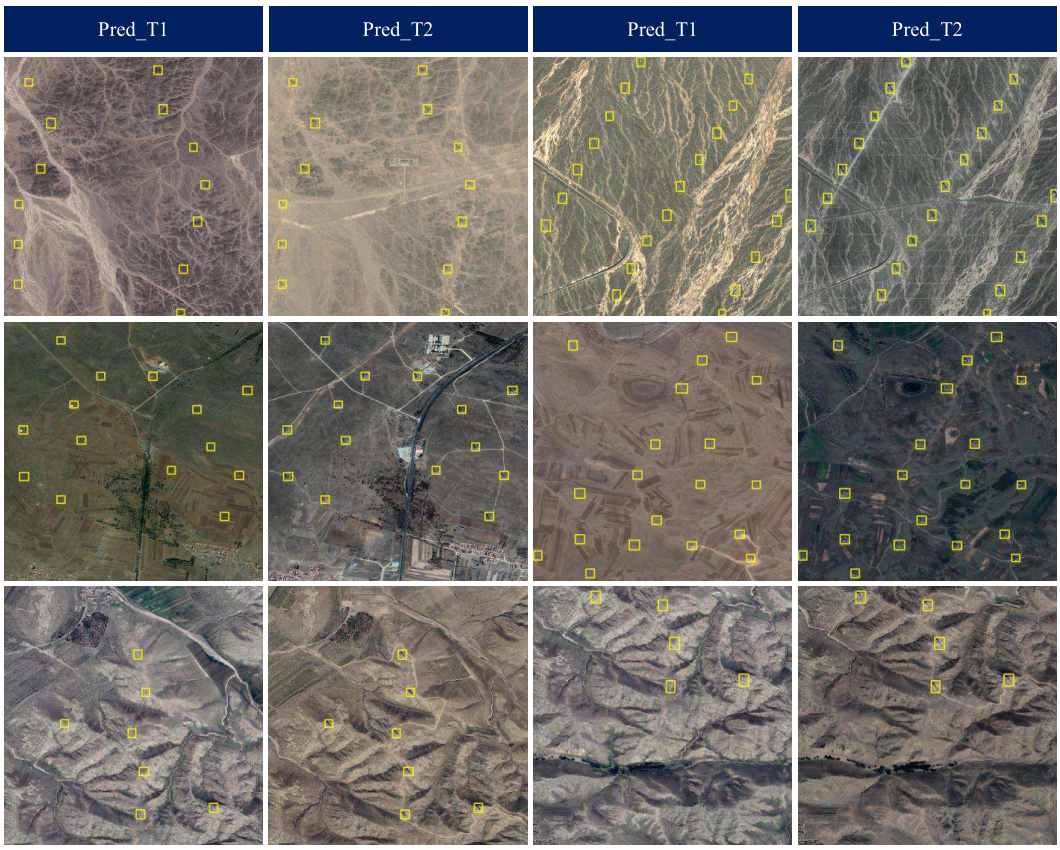


Fig. 14. Example of visualized detection results of the proposed SFDA-OCDet on the Wind turbine-CD dataset. (Pred_*T1*) and (Pred_*T2*) change detection results shown on *T1* and *T2* time-phase images.

TABLE V
COMPARISON OF THE ACCURACY OF EACH MODEL ON THE BUILDING-CD DATASET

Methods	Backbone	$AP_{0.5}$(%)	$AP_{0.5:0.95}$(%)
One-stage:			
SSD_cd	VGG16*	90.09	64.59
RetinaNet_cd	ResNet50*	85.55	62.91
YOLOv3_cd	DarkNet53*	88.44	59.68
FCOS_cd	ResNet50*	88.09	59.13
Yolact_cd	ResNet50*	84.50	59.65
QueryInst_cd	ResNet50*	84.91	60.99
Two-stage:			
Faster R-CNN_cd	ResNet50*	90.02	65.32
Mask R-CNN_cd	ResNet50*	90.01	65.33
Libra R-CNN_cd	ResNet50*	89.47	65.08
TridentNet_cd	ResNet50*	88.65	59.80
Dynamic R-CNN_cd	ResNet50*	89.75	66.42
Sparse R-CNN_cd	ResNet50*	84.50	61.24
SFDA-OCDet (Ours)	ResNet50*	**90.15**	**66.57**

* indicates the adoption of a Siamese structure for the backbone network.

of small object change detection task, and our proposed SFDA-OCDet can effectively detect the changes of small vehicles. Fig. 12 displays some results of SFDA-OCDet for vehicle change detection.

3) *Results on Building-CD Dataset:* The results of the comparison between SFDA-OCDet and other methods on the Building-CD dataset are shown in Table V. The buildings that change in the Building-CD dataset exhibit large-scale differences and dense distribution. The $AP_{0.5}$ and $AP_{0.5:0.95}$ of our proposed SFDA-OCDet on this dataset are 90.15% and 66.57%, respectively, which are higher than those of other one-stage and two-stage comparative methods, which verifies the validity of this method. The visualization results of SFDA-OCDet for building change detection are shown in Fig. 13.
4) *Results on Wind Turbine-CD Dataset:* As shown in the comparative experimental results of the methods in Table VI, the highest performance is obtained by the SFDA-OCDet proposed in this article. SFDA-OCDet achieves 98.67% and 69.72% on the $AP_{0.5}$ and $AP_{0.5:0.95}$ metrics, respectively, which are superior to other comparative methods. Fig. 14 presents the detection results of wind turbine changes for SFDA-OCDet.

F. Robustness Analysis

To further analyze the robustness of SFDA-OCDet, we conduct experiments to deeply explore its performance under challenging conditions such as seasonal variations, cloud

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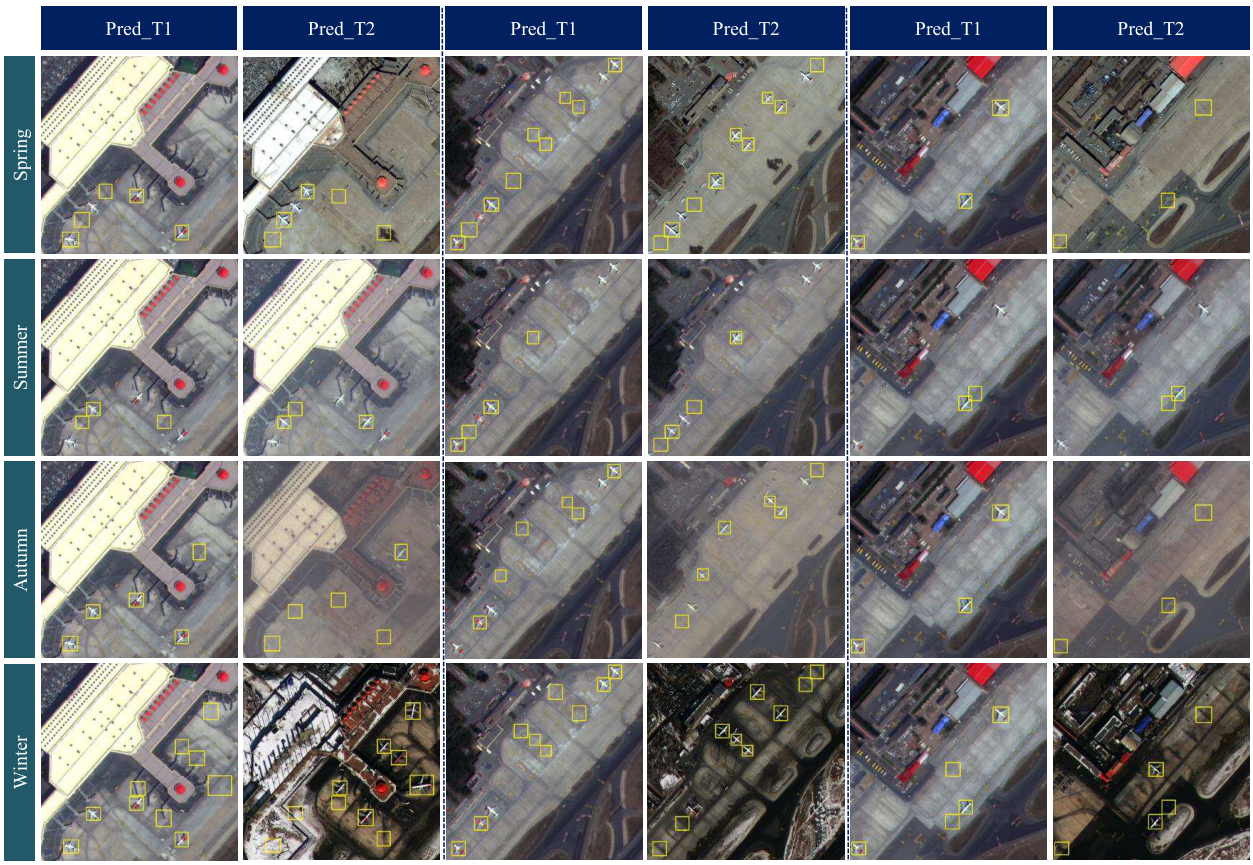


Fig. 15. Airplane change detection results under different seasons in spring, summer, autumn, and winter.

TABLE VI
COMPARISON OF THE ACCURACY OF EACH MODEL ON THE WIND TURBINE-CD DATASET

Methods	Backbone	$AP_{0.5}$(%)	$AP_{0.5:0.95}$(%)
One-stage:			
SSD_cd	VGG16*	97.46	64.73
RetinaNet_cd	ResNet50*	98.38	62.68
YOLOv3_cd	DarkNet53*	98.37	52.92
FCOS_cd	ResNet50*	98.49	54.45
Yolact_cd	ResNet50*	95.75	60.03
QueryInst_cd	ResNet50*	97.93	62.00
Two-stage:			
Faster R-CNN_cd	ResNet50*	97.79	63.90
Mask R-CNN_cd	ResNet50*	96.79	63.32
Libra R-CNN_cd	ResNet50*	97.62	64.93
TridentNet_cd	ResNet50*	97.53	55.09
Dynamic R-CNN_cd	ResNet50*	96.75	64.40
Sparse R-CNN_cd	ResNet50*	98.26	62.07
SFDA-OCDet (Ours)	ResNet50*	**98.67**	**69.72**

* indicates the adoption of a Siamese structure for the backbone network.

cover, sensor noise interference, illumination changes, and small object change detection.

1) Influence Analysis of Seasonal Variations: We use airplanes and wind turbines as study objects, and the images at the *T2* phase are collected from four different seasons, with the same dates for the *T1* phase images. Seasonal variations make differences in the lighting conditions and background characteristics of the images, especially in winter when the contrast produced by the snow background creates a greater disturbance for object-level change detection. Fig. 15 shows the results of airplane change detection for each of the three airport areas during the four seasons of spring, summer, autumn, and winter, where each row represents a season and each two columns correspond to a set of dual-temporal images. Fig. 16 shows the results of wind turbine change detection for each of the three wind farm areas during the four seasons. For the airport area, spring, summer, and autumn primarily affect the tone of the image, and winter adds a major difference in the snow background. For the wind farm area, the change of seasons changes the color tone of the image; the snowy background in winter weakens the features of the wind turbines, and the object features presented by the wind turbines and their shadows change. The above effects pose challenges for object-level change detection; however, our proposed SFDA-OCDet has better robustness to changes in seasons and can effectively detect changes in objects under different seasonal changes.

2) Influence Analysis of Cloud Cover: We acquire images of cloud-covered airplanes and wind turbines as *T2* phase data.

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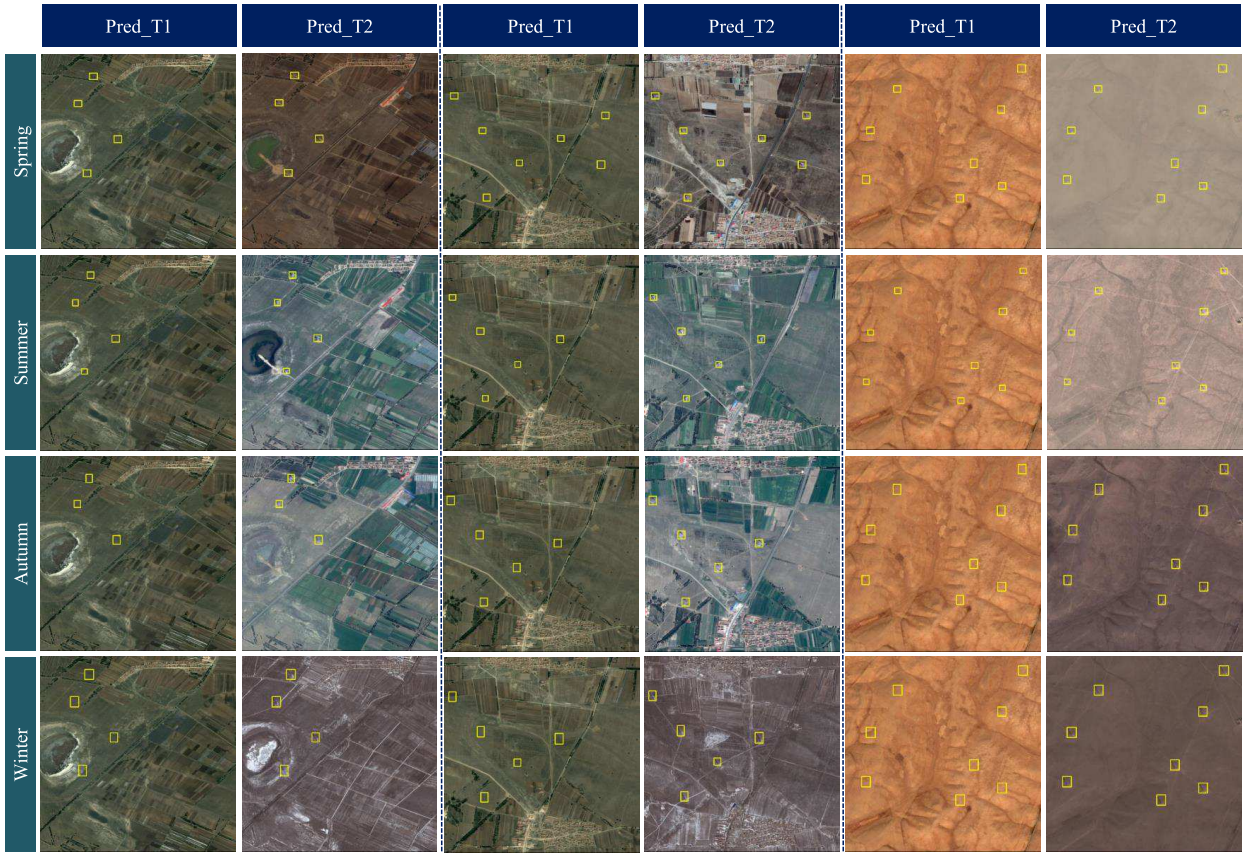


Fig. 16. Wind turbine change detection results under different seasons in spring, summer, autumn, and winter.

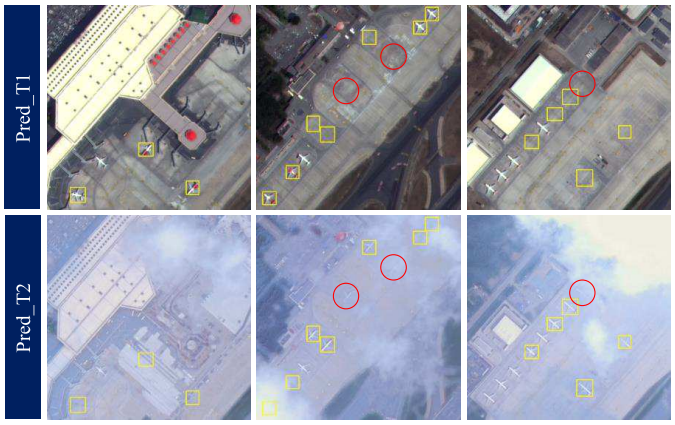


Fig. 17. Airplane change detection results in cloud cover scenarios.

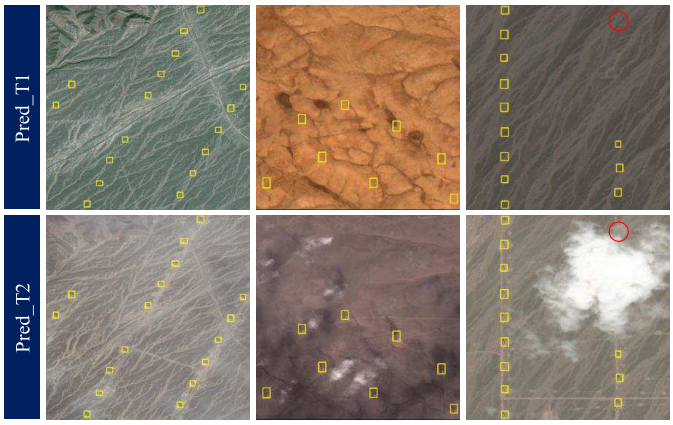


Fig. 18. Wind turbine change detection results in cloud cover scenarios.

In the cloud cover scenario, clouds reduce the clarity of the object and occlude the object. Figs. 17 and 18 present the results of airplane and wind turbine change detection in cloud cover scenarios, respectively, where each column represents the change detection results for a pair of dual-temporal images. For cloud cover conditions where airplanes and wind turbines become hazy and object features become faint, SFDA-OCDet is able to effectively handle and accurately identify changes in the objects. However, there are cases of failure when the SFDA-OCDet does not detect the presence of the object when the loss of airplane and wind turbine features due to cloud occlusion is so severe that it cannot present a full view of the object, as illustrated by the red circle in the figure below. Adding cloud-covered samples to the training set or combining it with a de-clouded algorithm can improve object-level change detection in scenarios with clouds.

3) Influence Analysis of Sensor Noise: To analyze the effect of sensor noise, we acquire a noisy airport image taken by a satellite on the one hand, and add Gaussian noise with five different parameters to all the *T2* phase images of the Airplane-CD test set on the other hand. We acquire a 2 m-resolution Gaofen-1 image covering Gimpo International

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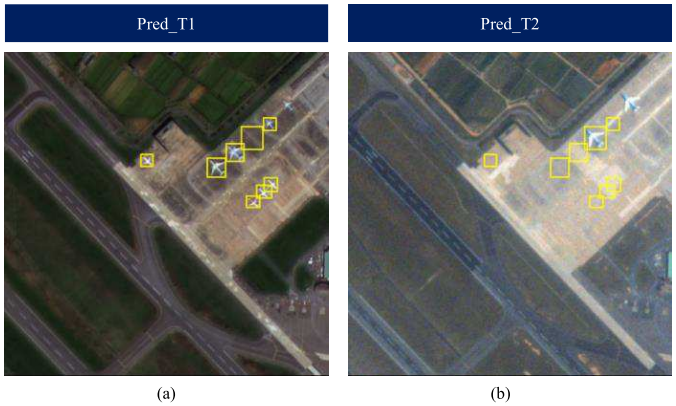


Fig. 19. Airplane change detection results in real sensor noise scenarios. (a) Gaofen-1 image. (b) Satellite image.

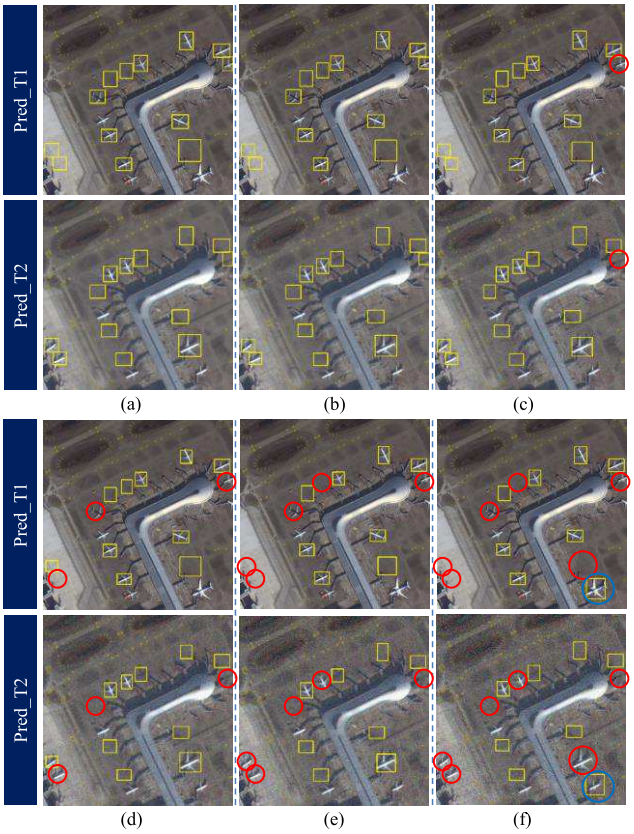


Fig. 20. Example of test sample change detection results before and after adding Gaussian noise. (a) Original test data. The variances of (b)–(f) are 0.001, 0.003, 0.005, 0.007, and 0.01, respectively. Red circle indicates missed detection, blue circle indicates false detection.

Airport in South Korea as *T1* phase data, and acquire a 2 m-resolution image of a satellite with sensor noise as *T2* phase data. The SFDA-OCDet we trained uses a dataset composed of 1 m-resolution Gaofen-2 and 0.65 m-resolution Gaofen-7 images and does not contain noise, so the predicted images shown in Fig. 19 not only have different data sources and different resolutions, but also include the interference of sensor noise. In this case, the SFDA-OCDet accurately detects the change in the airplanes.

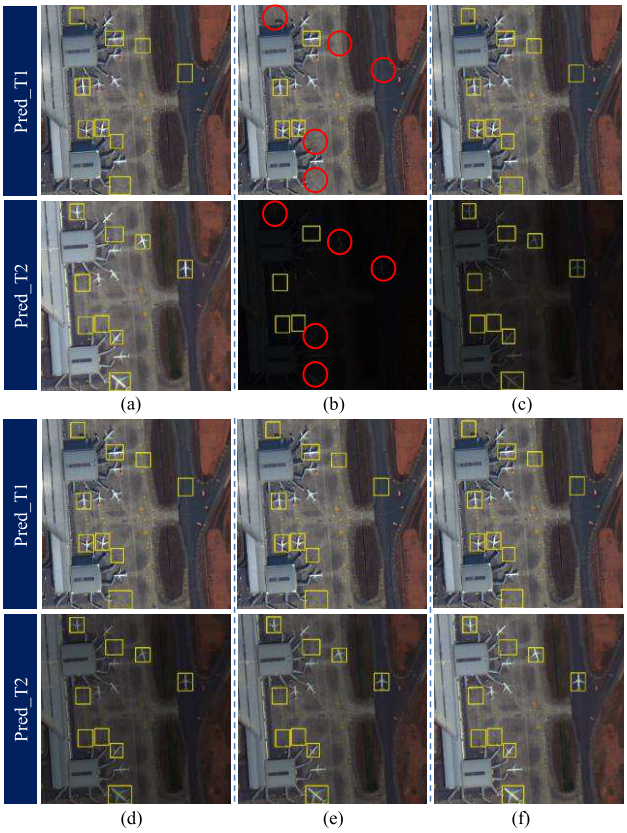


Fig. 21. Example of test sample change detection results for different darkening parameters. (a) Original test data. The parameter values of (b)–(f) are 0.1, 0.3, 0.5, 0.7, and 0.9, respectively. Red circle indicates missed detection.

TABLE VII
INFLUENCE OF DIFFERENT GAUSSIAN NOISE PARAMETERS ON THE DETECTION RESULTS OF AIRPLANE CHANGES

Gaussian noise variance value	$AP_{0.5}$(%)	$AP_{0.5:0.95}$(%)
—	98.60	78.33
0.001	96.60(↓2)	70.04(↓8.29)
0.003	94.36(↓4.24)	60.55(↓17.78)
0.005	88.34(↓10.26)	53.54(↓24.79)
0.007	83.01(↓15.59)	47.97(↓30.36)
0.01	74.35(↓24.25)	41.31(↓37.02)

We add Gaussian noise to all the images of the *T2* phase of the Airplane-CD test set, leaving the images of the *T1* phase unchanged. The mean value of the Gaussian noise is set to 0, and the variance is set to five different parameter values of 0.001, 0.003, 0.005, 0.007, and 0.01. Fig. 20 gives examples of airplane change detection results before and after the addition of Gaussian noise. Table VII shows the comparison of test set accuracy before and after adding noise, where the first row of results indicates the test accuracy without adding noise, and the rest indicate the test accuracy and degradation after adding noise. As the variance increases, the noise becomes progressively stronger, which affects the SFDA-OCDet in a progressively more pronounced way, and the test set accuracy

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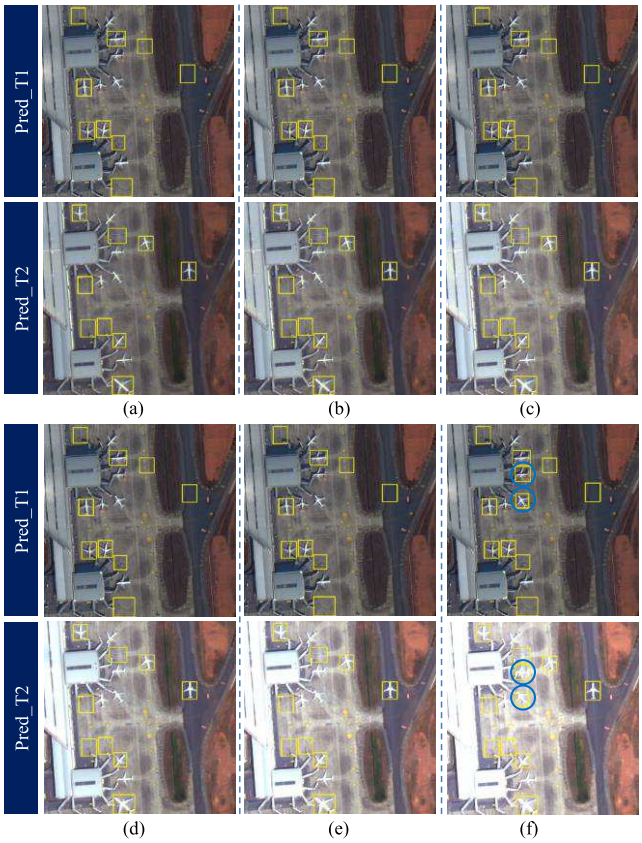


Fig. 22. Example of test sample change detection results for different brightening parameters. (a) Original test data. The parameter values of (b)–(f) are 1.1, 1.3, 1.5, 1.7, and 1.9, respectively. Blue circle indicates false detection.

gradually decreases. When the variance increases to 0.005, the $AP_{0.5}$ of SFDA-OCDet remains above 85%, indicating some robustness to Gaussian noise. Adding noisy images to the training data or combining the model with denoising algorithms can further improve the model's resistance to noise interference.

4) Influence Analysis of Illumination Changes: We previously analyzed the effect of seasonal variations on the performance of SFDA-OCDet. In addition to that, to further analyze the effect of illumination changes on the model performance, we perform brightness transformation on all the images of the *T2* phase of the Airplane-CD test set, and the *T1* phase images are kept unchanged. Our brightness transformation experiments consider both darkening and brightening, where five parameter values of 0.1, 0.3, 0.5, 0.7, and 0.9 are set for darkening and five parameter values of 1.1, 1.3, 1.5, 1.7, and 1.9 are set for brightening. The parameter value of 1 keeps the brightness unchanged from the original image. The smaller the parameter value, the darker the image; the larger the parameter value, the brighter the image. Table VIIIand Fig. 21 show the quantitative and visualization results for different darkening parameter values, respectively. The darker the image, the worse the SFDA-OCDet results. When the parameter value is between 0.3 and 0.9, SFDA-OCDet still maintains a high change detection accuracy with the $AP_{0.5}$ better than 90%. When the parameter value is set to 0.1, the performance decreases severely because the image is close to total black at this point, from which it is difficult to distinguish the target. Table IX and Fig. 22 show the quantitative and

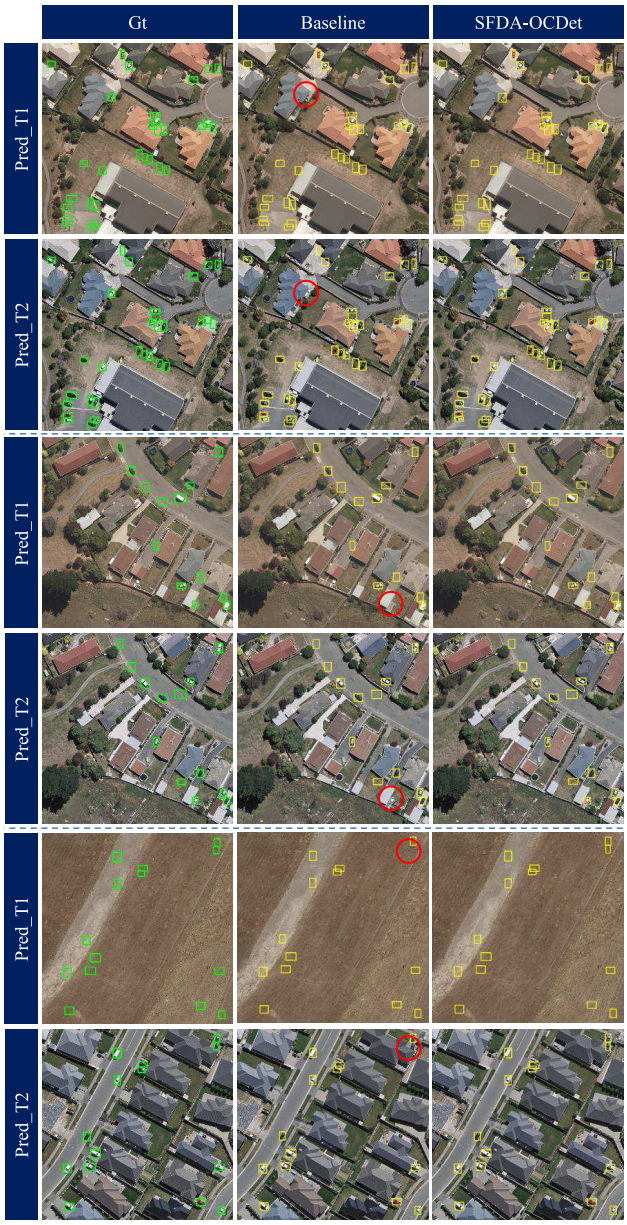


Fig. 23. Example of comparison of SFDA-OCDet and baseline's small vehicle change detection results. Red circle indicates missed detection.

TABLE VIII
INFLUENCE OF DIFFERENT DARKENING PARAMETER VALUES ON THE RESULTS OF AIRPLANE CHANGE DETECTION

brightness transformation parameter value	$AP_{0.5}$(%)	$AP_{0.5:0.95}$(%)
1	98.60	78.33
0.1	53.56(↓45.04)	38.91(↓39.42)
0.3	94.34(↓4.26)	72.41(↓5.92)
0.5	96.57(↓2.03)	75.71(↓2.62)
0.7	97.59(↓1.01)	77.30(↓1.03)
0.9	98.59(↓0.01)	78.05(↓0.28)

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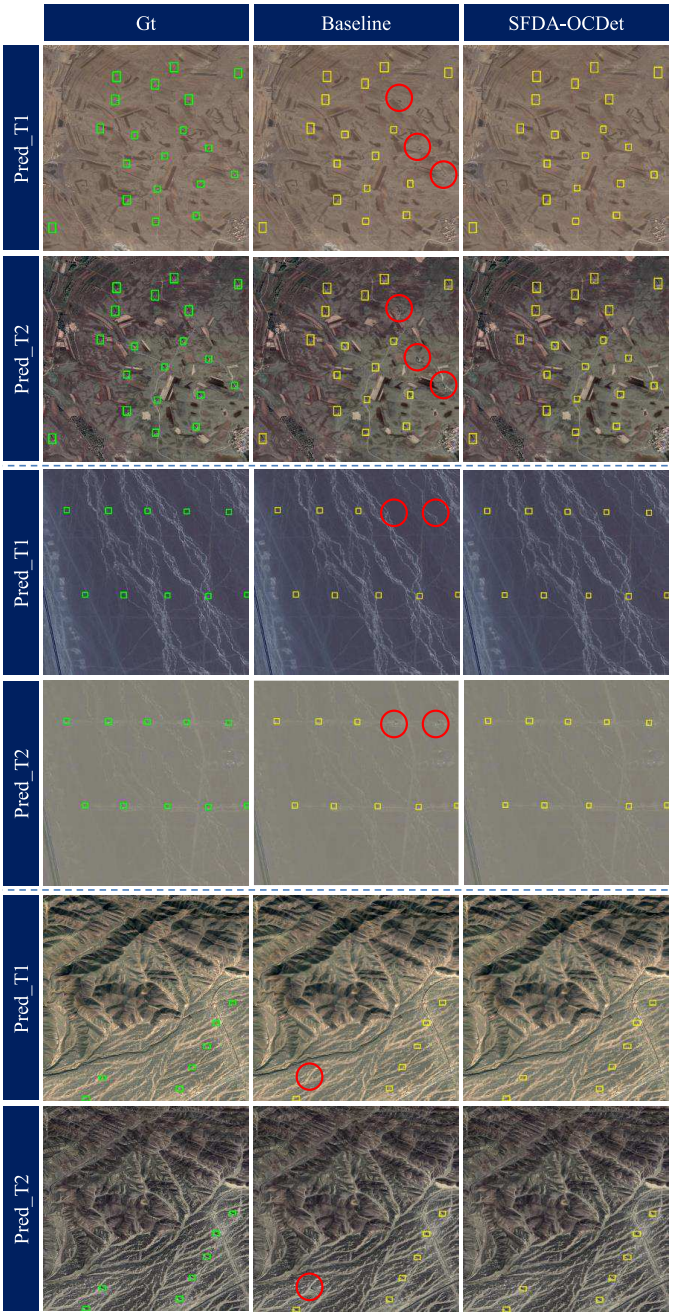


Fig. 24. Example of comparison of SFDA-OCDet and baseline's wind turbine change detection results. Red circle indicates missed detection.

visualization results for different brightening parameter values, respectively. The brighter the image is, the worse the effect of SFDA-OCDet is. When the parameter value is between 1.1 and 1.7, the $AP_{0.5}$ of SFDA-OCDet is better than 90% and stays at a high level. When the parameter value is set to 1.9, the performance drops more and more failures occur. The impact of brightness transformations on model performance can be reduced by adding darkened and lightened samples to the training set or by combining the model with brightness correction methods.

5) Analysis of Small Object Change Detection: Targets with an area less than 32×32 are defined as small objects according to the COCO definition. The objects included in the Vehicle-CD dataset and Wind turbine-CD dataset are all small objects. In addition, for the detection and change detection of wind turbines, it is more appropriate to select images with a resolution better than 1 m, as this can clearly display the target features. Therefore, the Wind turbine dataset constructed in this article also belongs to the low-resolution target change detection dataset. The objects in the two datasets are in complex environments, with vehicles distributed mainly in residential backgrounds where houses and trees cause interference, and wind turbines distributed in different complex backgrounds such as grasslands, desert, and mountains. Compared with the baseline, the SFDA-OCDet model we designed has improved the effect of small object change detection. The test results of the Vehicle-CD dataset show that SFDA-OCDet improved 0.58% and 1.26% over baseline in the $AP_{0.5}$ and $AP_{0.5:0.95}$ metrics, reaching 94.32% and 62.28%, respectively. The test results of Wind turbine-CD dataset show that SFDA-OCDet improves 1.05% and 4.79% over baseline in the $AP_{0.5}$ and $AP_{0.5:0.95}$ metrics, reaching 98.67% and 69.72%, respectively. Figs. 23 and 24, respectively, show the visual comparison results of the vehicle and the wind turbine. For the changes in the vehicles that are partially obscured by the buildings, the baseline model fails to detect them, while the SFDA-OCDet effectively detects this change. In low-resolution images, the wind turbine has a small geometric size and exhibits faintly blurred features. Moreover, the interference from various complex backgrounds causes the baseline model to fail to detect properly. SFDA-OCDet has improved the problem of missed detections and has better robustness for the detection of small object changes.

TABLE IX
INFLUENCE OF DIFFERENT BRIGHTENING PARAMETER VALUES ON THE RESULTS OF AIRPLANE CHANGE DETECTION

brightness transformation parameter value	$AP_{0.5}$(%)	$AP_{0.5:0.95}$(%)
1	98.60	78.33
1.1	98.54(↓0.06)	77.77(↓0.56)
1.3	98.43(↓0.17)	76.90(↓1.43)
1.5	97.56(↓1.04)	75.31(↓3.02)
1.7	93.98(↓4.62)	70.66(↓7.67)
1.9	87.51(↓11.09)	63.64(↓14.69)

G. Efficiency and Performance Trade-Off Analysis

To deeply explore the efficiency and performance of SFDA-OCDet, we carry out the analysis between different models, between different modules, and between different GPUs using the Airplane-CD dataset.

1) Analysis of Efficiency of Different Models: We compare the number of parameters and inference time of SFDA-OCDet with other methods, as shown in Table X. Our proposed SFDA-OCDet obtains the best performance on the four datasets at the cost of a slight increase in inference time. The inference time for an average pair of dual-temporal images

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TABLE X
COMPARISON OF PARAMETER NUMBER AND INFERENCE TIME OF EACH MODEL

Methods	Params (M)	FLOPs (G)	Inference Time (s)	FPS
One-stage:				
SSD_cd	24.39	173.98	0.0560	17.9
RetinaNet_cd	36.1	73.8	0.0525	19.0
YOLOv3_cd	61.52	86.91	0.0476	21.0
FCOS_cd	31.84	70.63	0.0499	20.0
Yolact_cd	32.36	52.13	0.0453	22.1
QueryInst_cd	106.0	55.48	0.0754	13.3
Two-stage:				
Faster R-CNN_cd	41.12	71.89	0.0557	18.0
Mask R-CNN_cd	41.12	71.89	0.0557	18.0
Libra R-CNN_cd	41.39	72.15	0.0571	17.5
TridentNet_cd	32.76	769.28	0.2172	4.6
Dynamic R-CNN_cd	41.12	71.89	0.0561	17.8
Sparse R-CNN_cd	106.0	55.48	0.0759	13.2
SFDA-OCDet (Ours)	53.97	74.35	0.0786	12.7

TABLE XI
COMPARISON OF EFFICIENCY AND PERFORMANCE OF DIFFERENT MODULES

Models	$AP_{0.5}$ (%)	$\Delta AP_{0.5}$	Inference Time (s)	$\Delta Time$ (s)
Baseline	96.81	—	0.0571	—
Baseline +SDA	97.72	0.91 (↑0.94%)	0.0600	0.0029 (↑5.08%)
Baseline +WFDA	97.75	0.94 (↑0.97%)	0.0652	0.0081 (↑14.19%)
Baseline +WBFP	97.75	0.94 (↑0.97%)	0.0647	0.0076 (↑13.31%)
SFDA-OCDet	98.6	1.79 (↑1.85%)	0.0786	0.0215 (↑37.65%)

increases by 0.0215 s compared to the baseline. However, SFDA-OCDet can still achieve frames per second (FPS) of 12.7, which exceeds ten FPS and fulfills the near real-time requirement [91], [92]. Moreover, unlike object detection, which processes 1 FPS, our method processes two frames at a time, which is strictly 25.4 FPS. The cost of performance improvement does not have a serious impact on real-time processing. Therefore, it is worthwhile to trade the slight time cost for the accuracy improvement.

2) Analysis of Efficiency of Different Improvement Modules: The improvement and percentage of $AP_{0.5}$ and the increase and percentage of inference time after adding the SDA module, WFDA module, and WBFP module to the baseline model are shown in Table XI, respectively. In addition to SFDA-OCDet, there is also the option of the baseline + SDA model, which not only brings a 0.94% improvement in $AP_{0.5}$, but also a smaller increase in inference time of only 5.08%.

TABLE XII
EFFICIENCY COMPARISON OF BASELINE AND SFDA-OCDET ON DIFFERENT GPUS

Item		GeForce RTX 3060	Quadro RTX 5000
Compute Capability		8.6	7.5
Inference Time (s)	Baseline	0.0571	0.1018
	SFDA-OCDet	0.0786	0.1239
FPS	Baseline	17.5	9.8
	SFDA-OCDet	12.7	8.1

3) Analysis of Efficiency of Different GPUs: We deploy the SFDA-OCDet and baseline trained on an NVIDIA GeForce RTX 3060 with a compute capability of 8.6 to an NVIDIA Quadro RTX 5000 graphics card with a compute capability of 7.5, and compare the inference time with the FPS, and the results are shown in Table XII. The decrease in compute capability of different GPUs leads to an increase in inference time and a decrease in FPS. On the NVIDIA Quadro RTX 5000, SFDA-OCDet's FPS is 8.1, close to near real-time speeds.

V. CONCLUSION

In this article, we propose a novel object-level change detection network for remote sensing images, called SFDA-OCDet. The proposed network contains three parts: Siamese backbone, spatial-frequency attention-weighted balanced feature neck, and change detection head. Specifically, in the backbone network portion, a pair of Siamese backbones with shared weights is used to extract dual-temporal image features. In the neck portion, the proposed SDA module and WFDA module are used to enhance the difference features of changed objects from spatial and frequency domains, respectively, and the designed WBFP module is used to reduce the feature information loss in the BFP process. In the head portion, the location and category information of the change object is output in the form of a bounding box. Adequate experimental results carried out on the four datasets validate the effectiveness of our proposed method. In future work, we will develop lightweight object-level change detection network to further improve inference efficiency. Meanwhile, we will construct an object-level change detection model under low-quality image conditions to improve the robustness under poor image conditions.

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